Optimal allocations with capacity constrained verification*

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Abstract

A principal has m identical objects to allocate among a group of n agents. Objects are desirable and the principal's value of assigning an object to a given agent is that agent's private information. The principal can verify up to k agents, where $k < m$, thereby perfectly learning their types. We show that when no monetary transfers are available, optimal mechanisms take a merit-with-guarantee form. Such mechanisms can be implemented in two stages: in the first stage, objects are allocated based on merit, with up to k agents verified; in the second stage, any remaining objects are distributed via a lottery that guarantees all low-reporting agents a positive allocation probability. This guarantee sustains truthful reporting and allows the mechanism to allocate efficiently to more agents than can be verified. Truthful reporting is a Bayes–Nash equilibrium but not an ex-post equilibrium.

Keywords: Mechanism Design with Evidence; Allocations; Verification

JEL classification: D82, D71

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1 Introduction

We examine an allocation problem with multiple identical objects. Agents possess private information about the principal's value of assigning them an object. Each agent desires at most one object, and monetary transfers are not available. Agents' private information is based on verifiable evidence, but the principal is constrained to verify only a fixed number of reports—fewer than the number of available objects. We study how the principal should optimally audit agents based on their reports and use the audit outcomes, together with agents' reports, to allocate objects. This problem differs from standard mechanism design due to the verifiability of private information, which arises in various economic contexts.

Many allocation problems share this structure. For example, public agencies often allocate goods on the basis of recipients' need. Such goods include social housing, energy subsidies, or other social benefits. Applicants' need levels—based on income, household composition, or other characteristics—are private but verifiable through documentation, consultations, or home visits. However, not all recipients can be verified, and many households necessarily receive benefits without thorough verification of their claims. A similar structure arises when goods are allocated on the basis of merit, such as research grants or work permits for skilled immigrants. Many countries operate points-based immigration systems where applicants' qualifications determine their eligibility for work permits or permanent residence (Miller 1999, Sweetman and Warman 2013). In these settings, more objects must be allocated than can be verified, and the principal often faces a capacity constraint on verification—a fixed number of caseworkers, auditors, or officers—rather than a constant marginal cost of verification.

To illustrate the environment we have in mind, consider the allocation of social housing in England. Local authorities maintain housing registers and allocate homes using a points-based system that ranks applicants by housing need, based on self-reported characteristics such as overcrowding, medical conditions, or risk of homelessness. Each application is assessed by a caseworker through desk-based checks, supporting documentation, and in some cases home visits. However, with long waiting lists and a limited number of caseworkers, not all claims can be thoroughly verified before allocation decisions are made. Among those who receive housing, only a subset are selected for detailed verification, with some cases selected at random for review. The setting thus exhibits the key features of our model: objects are allocated on the basis of privately known but verifiable characteristics, the number of objects allocated exceeds the number of verifications that can be conducted, and audits are conducted randomly among recipients. We analyze the optimal method for allocating m homes among n agents when at most $k < m$ agents can be verified.

To study this problem, we formulate a model of capacity-constrained verification in an allocation problem without transfers. We depart from the standard assumption that

verification is costly at the margin (Townsend 1979, Gale and Hellwig 1985, Ben-Porath, Dekel and Lipman 2014) and instead assume that the principal faces a fixed verification capacity. In the applications we have in mind, verification requires specialized human capital whose supply is fixed in the short run: caseworkers qualified to carry out home visits, trained immigration officers who can authenticate foreign credentials and conduct interviews, or academic experts with the narrow expertise needed to evaluate research proposals. Expanding this capacity requires recruiting and training new personnel, a process that operates on a much longer time scale than the time available before the allocation has to be finalized. A local housing authority assigning social housing cannot easily expand its team of caseworkers to match a larger pool of applicants with greater needs; a migration agency deciding how to allocate work permits this month cannot hire additional officers in response to a surge in high-quality applications; and a funding body reviewing grant applications cannot enlarge its expert panel after the call has closed. When the deadline for finalizing allocations is fixed before agents submit their reports, the number of audits that can be completed with the available personnel is effectively predetermined. Our model captures this short-run constraint on verification resources.

Before allocating objects, the principal may audit a subset of agents and learn their types perfectly. At most k agents can be audited. The principal obtains value $v(t_i)$ from assigning an object to agent i, and total payoff is the sum of values $v(t_i)$ across all assigned agents i. There are no monetary transfers and no direct costs of verification; the only constraint is the capacity limit on the number of checks. The mechanism design question is whom to check based on the submitted reports and how to allocate objects based on the reports and checks.

We find mechanisms that maximize the expected payoff from allocating m objects subject to k checks. The allocation rules of optimal mechanisms can be decomposed into two rules that operate sequentially: a merit-based rule in the first stage and a lottery-based rule in the second stage. The merit-based rule allocates objects "efficiently" to agents with the highest reported types, where the number of objects allocated in this way is governed by two thresholds. If fewer than k agents report a type above the lower threshold, all such agents receive an object and the remaining objects enter the second stage. If at least k agents report a type above the lower threshold, an agent receives an object in the first stage if either (i) his report is among the k highest, or (ii) his report is among the m highest and exceeds the higher threshold. Among agents who receive an object in the first stage, up to k are selected for verification, and any agent found lying receives no object. Agents who do not receive an object in the first stage and whose reports are below the higher threshold enter the second stage, where a lottery-based rule allocates the remaining objects so that all types below the higher threshold obtain an object in the lottery with the same probability.

To understand incentives for truthful reporting, note that the merit-based rule some-

times allocates more than k objects, so not all agents receiving an object in the first stage can be checked. This suggests that agents might be tempted to claim that they have a high type. The optimal mechanism mitigates these incentives by not always allocating all objects in the first stage: the lottery-based allocation of remaining objects ensures that even agents with low types retain a positive probability of receiving an object when reporting truthfully. In particular, in the mechanism we construct, their probability of obtaining an object through the lottery equals their probability of obtaining an object in the first stage without being audited when they falsely report a type above the higher threshold. However, consider a profile at which all reports exceed the higher threshold. At least one of the m recipients is not audited with certainty. If that agent's true type were instead below all other agents' types, truthful reporting would yield no object, whereas reporting his original high type would give him a positive chance of receiving one. Thus, while the optimal mechanism is Bayesian incentive compatible, it does not satisfy the stronger notion of ex-post incentive compatibility. This contrasts with models of costly verification, where optimal mechanisms can typically be implemented to satisfy these stronger incentive constraints.

1.1 Related Literature

The literature on costly state verification in principal-agent models was initiated by Townsend (1979). Our model differs from his, and the literature building on it (see e. g. Gale and Hellwig 1985, Border and Sobel 1987), since monetary transfers are not feasible in our model. In models without transfers, incentive constraints and economic trade-offs are different than in models with money.

Over the last years, there has been growing interest in models with state verification that do not allow for transfers, as initiated by Ben-Porath et al. (2014). The model studied by Ben-Porath et al. consists of a principal who wishes to allocate a single indivisible good among a group of agents, and each agent's type can be learned at a given cost. They find that the expected utility-maximizing mechanism for the principal is a favored-agent mechanism, where a predetermined favored agent receives the object unless another agent claims a value above a threshold, in which case the agent with the highest (net) type gets the object. Our model is similar in that there are no monetary transfers but differs in that our principal allocates several objects and can check a fixed number of agents at no cost.

Ben-Porath, Dekel and Lipman (2019) also consider the allocation of several identical objects. In the costly-verification version of their model, the principal allocates m objects and can verify agents at a constant marginal cost, and their results imply that a recursive favored-agent mechanism is optimal. The main difference from our model is that we assume a fixed audit capacity instead of a constant marginal cost. In many applications,

it seems natural that the cost of verification is convex in the number of audits, and the two models are polar cases of this specification. Having a fixed auditing capacity changes the nature of the optimal mechanism, and the incentive properties optimal mechanisms satisfy. We discuss why optimal mechanisms differ in the way they do in detail after stating our main result. With linear costs, the cost of an additional audit does not depend on how many other agents are audited, whereas with strictly convex costs it does, as in our model. We therefore expect some features of our optimal mechanism, such as its failure to be ex-post incentive compatible, to carry over to strictly convex costs whenever audits are used.

Other related papers on mechanism design with evidence include Glazer and Rubinstein (2004) and Glazer and Rubinstein (2006), which consider a situation in which an agent has private information about several characteristics and tries to persuade a principal to take a given action, and the principal can only check one of the agent's characteristics.[1] Other papers on optimal mechanisms in settings with evidence include Mylovanov and Zapechelnyuk (2017), Halac and Yared (2020), and Kattwinkel and Knoepfle (2023).

Methodologically, we build on results showing for which interim rules there exist corresponding ex-post rules. This approach was pioneered by Maskin and Riley (1984), who used it to characterize optimal auctions with risk-averse bidders. This question was first answered conclusively by Border (1991), who characterized the set of feasible interim allocation rules. More recently, Border's result was generalized by Che, Kim and Mierendorff (2013), who, among other things, allow for the allocation of m objects when agents face capacity constraints. We use a recent generalization by Valenzuela-Stookey (2024) that extends the characterization of feasible interim rules to settings with type-dependent ex-post constraints and that perfectly fits our setting.

The rest of the paper is organized as follows. In Section 2, we present the model, and characterize incentive compatible mechanisms. In Section 3, we describe the class of optimal mechanisms and discuss its main qualitative properties. In Section 4, we provide an argument for why it is optimal. In Section 5, we show how to determine the expected probability of winning at the lottery stage in the optimal mechanism and analyze the mechanism design problem for a large market. Section 6 concludes.

2 Model and preliminaries

A principal allocates m identical and indivisible objects to a set $N := \{1, \dots, n\}$ of agents, where $m < n$. Each agent i is characterized by a type $t \in T := [0, 1]$, which is private

[1] For additional papers on mechanism design with evidence, see also Green and Laffont (1986), Bull and Watson (2007), Deneckere and Severinov (2008), Ben-Porath and Lipman (2012), and Schweighofer-Kodritsch and Strausz (2024).

information to agent i. If agent i with type t obtains an object, agent i's utility is $u_i(t) > 0$ and the principal's utility from allocating to i is $v(t)$, where v is strictly increasing and satisfies $v(0) = 0$.[2] No agent needs more than one object. Types are drawn independently from a uniform distribution and we denote $\boldsymbol{t} = (t_i)_{i \in N}$, $\boldsymbol{t}_{-i} = (t_j)_{j \in N \setminus \{i\}}$, and $\boldsymbol{T} = [0,1]^N$.[3] The principal has a verification technology at his disposal and can learn the types of up to k agents perfectly, where $1 \leqslant k < m$ (we will use verification and audit interchangeably). The principal can commit to arbitrary mechanisms to allocate the objects, and we consider Bayes-Nash equilibria of the induced game.

We now define direct mechanisms for this setting. We capture stochastic mechanisms by using a random variable s, which is distributed independently of the types and uniformly on $[0,1]$ and whose realization is only observed by the principal. Formally, an *allocation rule* is a profile of functions $p_i : \boldsymbol{T} \times [0,1] \to \{0,1\}$ satisfying, for all $\boldsymbol{t}$ and s, $\sum_{i=1}^n p_i(\boldsymbol{t}, s) \leqslant m$ and an *audit rule* is a profile of functions $a_i : \boldsymbol{T} \times [0,1] \to \{0,1\}$ satisfying, for all $\boldsymbol{t}$ and s, $\sum_{i=1}^n a_i(\boldsymbol{t}, s) \leqslant k$.[4]

A *direct mechanism* (p, a) consists of an allocation rule and an audit rule. Given a profile of reports $\boldsymbol{t}$ and a realization s of the random variable, agent i is audited if $a_i(t_i, \boldsymbol{t}_{-i}, s) = 1$. We say that a *lie by agent i is detected* if i's true type t_i' differs from his report t_i and $a_i(t_i, \boldsymbol{t}_{-i}, s) = 1$. Agent i obtains an object if no lie by agent i was detected and $p_i(t_i, \boldsymbol{t}_{-i}, s) = 1$, and obtains no object otherwise.

Using arguments similar to those in Ben-Porath et al. (2014) and Erlanson and Kleiner (2020), one can show that it is without loss of generality to restrict attention to direct mechanisms that are Bayesian incentive compatible (see Appendix D for details).

Definition 1. *A mechanism (p, a) is Bayesian incentive compatible (BIC) if, for all $i \in N$ and all $t, t' \in T$*

$$u_i(t') \cdot \mathbb{E}_{\boldsymbol{t}_{-i}, s}[p_i(t', \boldsymbol{t}_{-i}, s)] \geqslant u_i(t') \cdot \mathbb{E}_{\boldsymbol{t}_{-i}, s}[p_i(t, \boldsymbol{t}_{-i}, s)(1 - a_i(t, \boldsymbol{t}_{-i}, s))]. \tag{1}$$

The principal's problem can formally be stated as:

$$\max_{p,a} \mathbb{E}_{\boldsymbol{t}, s}\left[\sum_{i \in N} p_i(t_i, \boldsymbol{t}_{-i}, s) v(t_i)\right]$$
$$\text{s.t. } (p, a) \text{ is BIC,}$$

[2]Since the principal's payoff from not allocating an object is zero, the assumption $v(0) = 0$ is not only a normalization. Our results extend to $v(0) > 0$ because the mechanisms we construct allocate all m objects; however, if $v(0) < 0$, the principal may prefer to leave objects unallocated, which we do not consider.

[3]Our analysis also covers i.i.d. types with a continuous and strictly increasing CDF F. In this case, we identify each type with its quantile, which is uniformly distributed on $[0,1]$, and interpret $v(t)$ as the principal's value from allocating to an agent whose type has quantile t. Since F is strictly increasing, an audit reveals an agent's quantile if and only if it reveals his type, so incentive constraints are unaffected.

[4]We drop the argument s for allocation/audit rules that are deterministic for almost every $\boldsymbol{t}$.

and we say a mechanism is *optimal* if it solves this problem.

Reformulating the principal's problem By symmetry, it is without loss of optimality to restrict attention to symmetric mechanisms that treat all agents identically, and henceforth we will do that.[5] It will also be useful to formulate this mechanism design problem in terms of interim rules. Given a mechanism (p, a), its *interim rules* are defined for all $t \in T$ by

$$P(t) := \mathbb{E}_{\boldsymbol{t}_{-i},s}[p_i(t, \boldsymbol{t}_{-i}, s)]$$
$$A(t) := \mathbb{E}_{\boldsymbol{t}_{-i},s}[a_i(t, \boldsymbol{t}_{-i}, s)].$$

We say that p *induces* P and a *induces* A.

We can then simplify the Bayesian incentive compatibility constraints. First, since $u_i(t') > 0$ for all $i \in N$ and all $t' \in T$, $u_i(t')$ cancels from both sides of equation (1). Hence, agents' utilities matter only through their sign: every type strictly prefers obtaining an object, so each agent simply maximizes his probability of obtaining one. Thus, a mechanism is BIC if and only if, for all $t, t' \in T$,

$$\mathbb{E}_{\boldsymbol{t}_{-i},s}[p_i(t', \boldsymbol{t}_{-i}, s)] \geqslant \mathbb{E}_{\boldsymbol{t}_{-i},s}[p_i(t, \boldsymbol{t}_{-i}, s)(1 - a_i(t, \boldsymbol{t}_{-i}, s))] \tag{2}$$

Note that, to ensure that agent i reports truthfully, it is only helpful to audit agent i if $p_i(t, \boldsymbol{t}_{-i}, s) = 1$. Otherwise, the audit has no impact on the expected probability of being allocated an object when reporting a lie. Using this insight, we can further simplify the incentive constraints expressed in (2) by correlating the allocation and audit rule. After the random variable s is drawn, agent i is either *scheduled* to get an object, $p_i(\boldsymbol{t}, s) = 1$, or not, $p_i(\boldsymbol{t}, s) = 0$. Since we cannot penalize an agent who is not scheduled to get an object, we can restrict attention to auditing only agents who are scheduled to get an object. Therefore, without loss of generality we can assume that $a_i(\boldsymbol{t}, s) = 1$ only if agent i is scheduled to get an object $p_i(\boldsymbol{t}, s) = 1$. Henceforth, we restrict attention to mechanisms satisfying this. Using this observation, $\mathbb{E}_{\boldsymbol{t}_{-i},s}[p_i(t, \boldsymbol{t}_{-i}, s)a_i(t, \boldsymbol{t}_{-i}, s)] = \mathbb{E}_{\boldsymbol{t}_{-i},s}[a_i(t, \boldsymbol{t}_{-i}, s)]$ and the right-hand side of (2) simplifies to $P(t) - A(t)$.

These observations imply the following characterization of Bayesian incentive compatibility.

Lemma 1. *A symmetric mechanism (p, a) is BIC if and only if its interim rules (P, A) satisfy, for all $t \in T$,*

$$A(t) \geqslant P(t) - \inf_{t' \in T} P(t'). \tag{3}$$

[5] That is, we focus in the following on mechanisms satisfying, for all $\boldsymbol{t}$ and permutations σ, $\mathbb{E}_s[p_i(\boldsymbol{t}, s)] = \mathbb{E}_s[p_{\sigma(i)}(\boldsymbol{t}_\sigma, s)]$, and similarly for a and pa.

3 Optimal mechanisms

In this section, we define the class of merit-with-guarantee mechanisms and show that an optimal mechanism can always be found in this class.

3.1 Merit-with-guarantee mechanisms

We begin by discussing some benchmark allocation rules. The first-best rule allocates objects to the agents with the m highest types. However, if only $k < m$ agents can be verified, this rule cannot be part of an incentive compatible mechanism: an agent with the lowest type would receive an object with probability zero by reporting truthfully, but with positive probability by reporting a higher type because not every agent receiving an object can be verified.

A simple incentive compatible alternative is to allocate objects only to the k highest reports, verifying these reports, and allocating the objects if the reports are truthful. The remaining $m - k$ objects are then allocated randomly among the remaining agents. Incentive constraints are slack in this mechanism: a misreport can increase an agent's probability of obtaining an object only if it is among the k highest reports, in which case it is verified with probability one and yields zero probability of receiving an object, whereas truthful reporting yields a strictly positive probability of receiving an object.

Since incentive constraints are slack, the principal can improve upon this mechanism by sometimes allocating more than k objects efficiently while verifying only some of those reports. One way to do so is to introduce a cutoff and allocate objects to the agents with the m highest reports whenever these reports exceed the cutoff, verifying k of these reports, and allocating any remaining objects randomly below the cutoff. This mechanism is incentive compatible for sufficiently high cutoffs. The optimal mechanism shares this feature, but introduces an intermediate interval in which only the k highest reports obtain an object. These reports are verified with certainty, so the intermediate interval uses verification capacity that would otherwise remain idle when few reports exceed the high cutoff.

Definition 2. *A mechanism (p, a) is a* merit-with-guarantee mechanism *if there exist thresholds $0 \leqslant x_k \leqslant x_m \leqslant 1$ and a constant $\varphi \in (0, \frac{m}{n}]$ such that, for the three regions $T^\ell := [0, x_k]$, $T^k := (x_k, x_m]$, and $T^m := (x_m, 1]$, the allocation rule p can be decomposed as $p = p^M + p^\ell$, where:*

(i) $p_i^M(\boldsymbol{t}) = 1$ if either (a) $t_i > x_m$ and t_i is among the m highest reports in $\boldsymbol{t}$, or (b) $t_i > x_k$ and t_i is among the k highest reports in $\boldsymbol{t}$; otherwise $p_i^M(\boldsymbol{t}) = 0$;[6]

[6] If $t_i = t_j$ for some i, j, we use an arbitrary tie-breaking rule.

(ii) p^ℓ *allocates objects only to agents who do not obtain an object under* p^M *and whose reports lie in* $T^\ell \cup T^k$*; moreover,* $\mathbb{E}_{\boldsymbol{t}_{-i},s}[p_i^\ell(t_i, \boldsymbol{t}_{-i}, s)] = \varphi$ *for all* i *and all* $t_i \in T^\ell \cup T^k$*; and*

(iii) the audit rule a *randomly selects up to* k *agents so that*

$$\mathbb{E}_{\boldsymbol{t}_{-i},s}[a_i(t_i, \boldsymbol{t}_{-i}, s)] \geqslant \mathbb{E}_{\boldsymbol{t}_{-i},s}[p_i(t_i, \boldsymbol{t}_{-i}, s)] - \varphi$$

for all i *and all* $t_i \in T$*.*

Any merit-with-guarantee mechanism can be implemented in two stages: In the first stage, objects are scheduled to be allocated according to the *merit-based allocation rule* p^M: those agents whose reports are among the m highest reports and lie in T^m and those whose reports are among the k highest reports and lie in T^k are scheduled to obtain an object; some of those agents are chosen by the audit rule for a verification; if their reports are truthful (which they will be in equilibrium), the agents that are scheduled for allocation obtain their object (in case of a lie being detected, the lying agent does not receive an object). Objects not allocated in this first stage enter the second-stage lottery. All agents with reports in T^ℓ, as well as agents with reports in T^k who are not scheduled for allocation in the first stage, participate in this lottery. In this second stage, the *lottery-based allocation rule* p^ℓ allocates some of the remaining objects so that all agents with types in T^ℓ and T^k have the same expected probability φ of receiving an object in the lottery stage.

To satisfy property (iii), it is sufficient for the optimal merit-with-guarantee mechanism to only audit agents who are scheduled for allocation in the first stage, i.e., agents for whom $p_i^M(\boldsymbol{t}) = 1$. In that case, low reports that lie in T^ℓ will never be audited. Moreover, all agents with reports in T^k who are scheduled to receive an object in the first stage are audited for sure: Since $p_i = p_i^M + p_i^\ell$ and $\mathbb{E}_{\boldsymbol{t}_{-i},s}[p_i^\ell(t_i, \boldsymbol{t}_{-i}, s)] = \varphi$ by (ii), property (iii) implies $\mathbb{E}_{\boldsymbol{t}_{-i},s}[a_i(t_i, \boldsymbol{t}_{-i}, s)] \geqslant \mathbb{E}_{\boldsymbol{t}_{-i}}[p_i^M(t_i, \boldsymbol{t}_{-i})]$. And since only agents scheduled for first-stage allocation are audited, we also have $a_i(t_i, \boldsymbol{t}_{-i}, s) \leqslant p_i^M(t_i, \boldsymbol{t}_{-i})$, yielding $a_i(t_i, \boldsymbol{t}_{-i}, s) = p_i^M(t_i, \boldsymbol{t}_{-i})$. On the other hand, because up to m objects are scheduled for allocation to types in T^m and at most k of the reports can be verified, the merit-with-guarantee mechanism has to carefully balance incentives for low types to be truthful. We describe incentives in merit-with-guarantee mechanisms in more detail below.

Figure 1 illustrates a merit-with-guarantee mechanism. The principal allocates objects efficiently within the T^m interval. Only if fewer than m agents have reports in T^m does the principal allocate objects in the first stage to agents with reports in T^k who are among the k highest reports. Any remaining objects enter the lottery stage and are allocated among agents with reports in T^ℓ and T^k who did not obtain an object in the first stage. Example 1 illustrates how the merit-with-guarantee mechanism allocates objects and determines audits.

Example 1. *Suppose that three objects and two audits are available ($m = 3,\ k = 2$).*

If at least three reports are in T^m, the audit rule randomly selects two of the three highest reports for verification. If no lie is detected, all three objects are allocated to the agents with the three highest reports, and no object remains for the second-stage lottery.

Suppose instead that exactly one report is in T^m and at least one report is in T^k. Then the report in T^m and the highest report in T^k are verified, and—if no lie is detected—both agents receive an object. The remaining object is allocated in the second stage via a lottery among all unassigned agents with reports in T^ℓ and T^k.

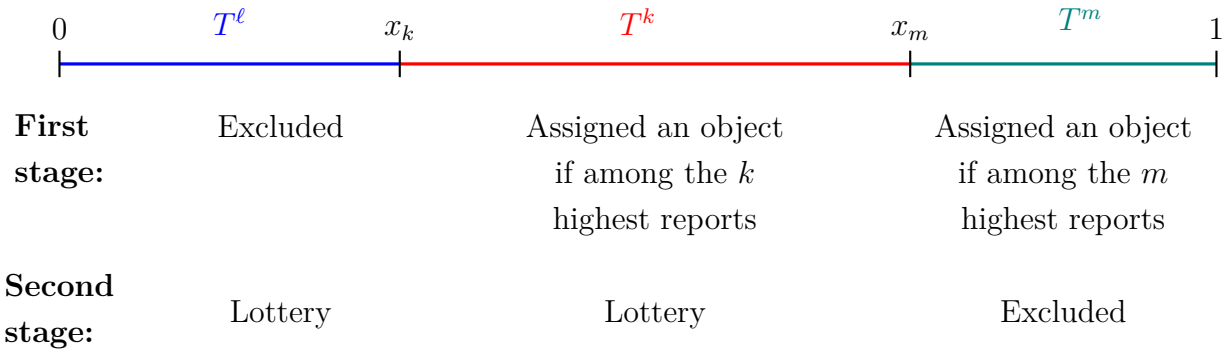


Figure 1: Allocation rule in a merit-with-guarantee mechanism.

3.2 A merit-with-guarantee mechanism is optimal

The following result shows that a merit-with-guarantee mechanism is optimal.

Theorem 1. *A merit-with-guarantee mechanism is optimal. Moreover, every optimal and symmetric mechanism equals a merit-with-guarantee mechanism almost surely.*

A key feature of a merit-with-guarantee mechanism is that even low types obtain an object with some probability $\varphi > 0$, which we call the *guarantee* of the mechanism. To understand incentives for truthful reporting, consider an agent with a type $t_i \in T^\ell$. Under truthful reporting, this agent obtains an object only in the lottery stage, which happens with probability φ.

If the agent misreports as a type in T^k, he is scheduled to receive an object in the first stage only if his report is among the k highest. In that case, the report is audited with probability one, and the agent receives no object because the lie will be detected. Since the lottery probability is equal to φ for all reports in T^ℓ and T^k, such a deviation is not profitable.

If the agent instead misreports as a type in T^m, he may obtain an object in the first stage if more than k agents report types in T^m, since not all such reports can be verified and the lie might not be detected. However, agents reporting in T^m do not enter

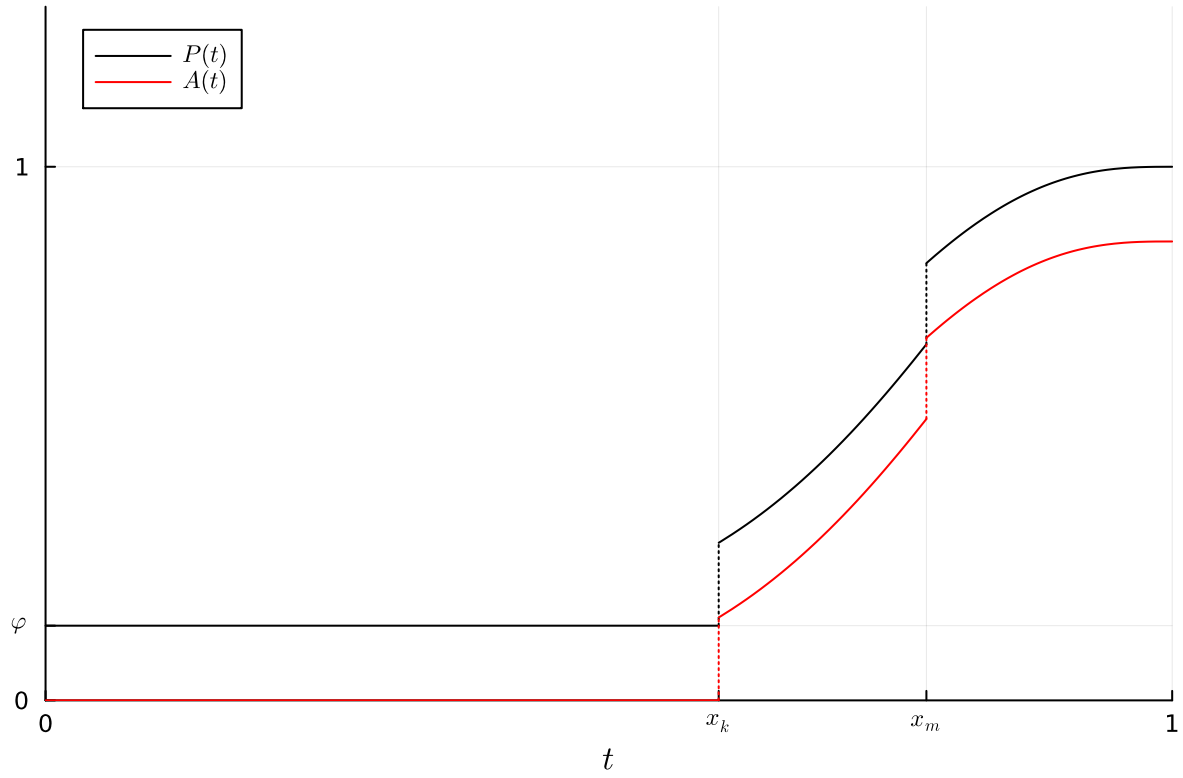


Figure 2: This figure illustrates the induced interim allocation and audit rules from the optimal merit-with-guarantee mechanism in an example where $n = 8, m = 3, k = 2$ and $\varphi = 0.14$.

the lottery stage, so the agent gives up on the probability of obtaining an object in the lottery stage. Provided that T^m is not too large, the probability that more than k agents report in T^m is small. Therefore, an agent who sends a misreport that is in T^m faces a high risk that his lie is detected; taking expectations over the other agents' reports, it is therefore optimal for type t_i to report truthfully.

While it is optimal, in expectation over the others' reports, for each agent to report truthfully, this is not necessarily optimal if an agent knew the others' reports. To see this, consider a profile $\boldsymbol{t}$ of distinct reports in T^m. The m highest reports receive an object, and since not all m recipients can be audited, at least one of them, say agent i, is not audited with certainty. Now suppose that agent i's true type is $t_i' < \min_{j \neq i} t_j$, while the other agents' types are given by $\boldsymbol{t}_{-i}$. If all agents are truthful, the m highest reports receive an object; since agent i's report is not among the m highest ones, he receives no object. If, instead, he misreports as t_i, the report profile is $\boldsymbol{t}$ and he receives an object whenever he is not audited. This event has positive probability, so the deviation is profitable. Thus, the merit-with-guarantee mechanism is generally not ex-post incentive compatible. In fact, this is not just a feature of the specific merit-with-guarantee mechanism: no optimal

mechanism, symmetric or not, satisfies these stronger incentive constraints.[7]

Remark 1. *No optimal mechanism satisfies ex-post incentive constraints.*

The merit-based allocation rule is nonetheless monotone: whenever an agent receives an object in the first stage, so does every agent with a higher report. Allocative inefficiencies therefore arise only because the objects that are not allocated in the first stage are allocated by lottery. Although agents with reports in T^m do not participate in the lottery, the overall interim allocation rule is also nondecreasing: an agent with a report in $T^\ell \cup T^k$ obtains an object only if fewer than m other agents report types in T^m, and in that case any report in T^m obtains an object in the first stage.

To gain intuition for why a merit-with-guarantee mechanism achieves a high payoff for the principal, recall that allocating objects only to the k highest types and distributing the remaining $m-k$ objects randomly leaves incentive constraints slack. A merit-with-guarantee mechanism improves upon this by sometimes allocating more than k objects to high reports—specifically, by allocating up to m objects when reports are in T^m. Incentive compatibility is preserved as long as the probability that more than k agents report types in T^m is sufficiently small.

In the optimal merit-with-guarantee mechanism, the region T^m cannot be expanded further without violating incentive constraints for low types. At the same time, when fewer than k objects are allocated to agents in T^m, some verification capacity remains unused. The optimal mechanism addresses this by allocating some of the remaining objects to agents with reports in T^k and auditing these reports with probability one. This intuition for optimality is incomplete but captures some insights into why such mechanisms are optimal. Taking the perspective of interim rules in Section 4 will provide more detailed insights into why this mechanism is optimal and how the thresholds x_k and x_m are determined.

Theorem 1 shows that an optimal mechanism can always be found in the class of merit-with-guarantee mechanisms, and that among symmetric mechanisms only merit-with-guarantee mechanisms can be optimal. Note that the definition of a merit-with-guarantee mechanism does not fully specify the allocation and audit rules; for the lottery-based allocation, it only specifies that the probability of obtaining an object in the lottery stage is constant across types in $T^\ell \cup T^k$. One reason why we leave the exact implementation unspecified is that, once the guarantee and the thresholds are chosen optimally, any lottery-based allocation that induces the required constant interim rules is optimal. If we

[7]By Theorem 1, every optimal symmetric mechanism is a merit-with-guarantee mechanism. Its guarantee is below $\frac{m}{n}$, hence T^m is nonempty (see Section 5.1), and the argument above applies. If an asymmetric optimal mechanism were ex-post incentive compatible, randomly relabeling the agents would yield a symmetric optimal mechanism that is ex-post incentive compatible as well.

want to discuss whether a given mechanism is optimal, this is the only implication we can draw from our analysis about the lottery-based allocation. In this sense, Theorem 1 captures all the qualitative insights we can gain into optimal mechanisms.

Another reason why we leave the exact implementation unspecified is that typically there are many specifications and there does not appear to be a canonical one. A natural approach would be to allocate all objects available in the lottery stage uniformly at random among the agents entering the lottery. While this often yields winning probabilities that are almost constant, this does not work precisely in general. The reason is that because the probability of receiving an object in the merit stage is increasing in the type for types in T^k, their probability of entering the lottery is decreasing in their report. Moreover, the expected number of objects entering the lottery stage conditional on an agent with type $t \in T^k$ entering the lottery generally depends on t. This is because the event that the agent does not receive an object in the merit stage conveys information about the other agents' reports, and hence about the number of objects allocated in that stage. Therefore, to maintain the same interim probability φ, the conditional probability of winning the lottery cannot be uniform across agents in general. Our proof establishes that one can always implement the lottery in a way that yields constant winning probabilities of φ.[8]

As mentioned in the introduction, Ben-Porath et al. (2019) show that a recursive favored-agent mechanism is optimal if audits have a constant marginal cost. In our symmetric setting, and with the favored agents chosen at random, its allocation rule is that of a merit-with-guarantee mechanism with $x_k = x_m$: the m highest reports above a single cutoff obtain an object, and any remaining objects are allocated by lottery among agents with reports below the cutoff. The intermediate region T^k is absent in their mechanism because there an audit costs the same at every report profile, whereas in our model an audit is free as long as fewer than k agents are audited and impossible otherwise. At profiles at which fewer than k agents report types in T^m, audits that would otherwise remain idle are therefore used to allocate objects to the highest reports in T^k, which are audited with certainty. Thus, a report has to exceed only x_k to obtain an object on merit if it is among the k highest reports, so that an audit is available for it, and has to exceed x_m otherwise.

The guarantee also plays a different role in the two models. In both models, a higher guarantee lowers the audit probability required to support a given allocation probability (Lemma 1). Under a constant marginal cost, this saves audit costs. In our model, a higher

[8]One way to construct an ex-post allocation rule p that induces the optimal interim allocation rule P follows from observing that P is a mixture of the extreme points characterized in Kleiner, Moldovanu and Strack (2021). Following the construction in the proof of their Theorem 3, one obtains a particular ex-post allocation rule that is a mixture of ex-post allocation rules that contain pooling intervals (an analogous argument is used by Maskin and Riley (1984) for the single-good case to prove their Theorem 7). We illustrate another explicit construction in the context of Example 2.

guarantee instead allows the principal to allocate more than k objects on merit when many agents report high types, even though some of these agents then remain unaudited. The optimal guarantee trades off this benefit against allocating more objects at random to low types (see Section 5.1).

That some merit winners remain unaudited also explains why optimal mechanisms in our model cannot satisfy ex-post incentive constraints (Remark 1). By contrast, under a constant marginal cost, the principal can deter misreports at the lowest expected cost by auditing an agent who obtains an object at exactly those profiles at which he would not have obtained it with the lowest report; with this audit rule, their mechanism is ex-post incentive compatible.

4 Why a merit-with-guarantee mechanism is optimal

In this section, we explain in more detail why a merit-with-guarantee mechanism is optimal. Along the way, we will obtain additional insights into the structure of optimal mechanisms and show how an optimal merit-with-guarantee mechanism can be found by optimizing over a single parameter. To do all this, it will be useful to formulate the problem in terms of interim rules because this significantly reduces its dimensionality. We will see that the resource constraints imply that any interim rules must satisfy specific conditions, which in turn determine the shape of the various allocation regions in a merit-with-guarantee mechanism. We formulate an upper bound on the principal's payoff and show that a particular merit-with-guarantee mechanism might attain this bound.

In the following, we define which audit rules are feasible. Recall our convention that in any mechanism $a_i(\boldsymbol{t}, s) = 1$ only if $p_i(\boldsymbol{t}, s) = 1$, which is without loss of generality and which we used to simplify incentive constraints.

Definition 3. *An audit rule a is* feasible given an allocation rule p *if $a_i(\boldsymbol{t}, s) \leqslant p_i(\boldsymbol{t}, s)$ for all $i \in N$, $\boldsymbol{t} \in \boldsymbol{T}$, and $s \in [0, 1]$. A pair of interim rules (P, A) is* feasible *if there exist an allocation rule p and an audit rule a which is feasible given p that induce (P, A).*

We can now reformulate the principal's problem using interim rules as follows:

$$\max_{P,A} \; n\,\mathbb{E}_t[P(t)v(t)] \qquad \text{(Opt)}$$

$$\text{s.t. } (P, A) \text{ is feasible and}$$

$$A(t) \geqslant P(t) - \inf_{t' \in T} P(t') \text{ for all } t \in T.$$

To find the optimal interim rules, we start by deriving constraints that any feasible mechanism must satisfy. These constraints are due to the limited availability of objects and audits (for related conditions, see Border 1991).

A feasibility constraint on any interim allocation rule P, which stems from at most m objects being available, is that for any type t,

$$n \int_t^1 P(x)\,\mathrm{d}x \leqslant c^{allo}(t) \tag{4}$$

where

$$c^{allo}(t) := \sum_{i=1}^{n} \min\{i, m\} \binom{n}{i} (1-t)^i t^{n-i}.$$

To see this, note that the left-hand side of (4) denotes the expected number of agents with a type above t that are assigned an object. For any $i \in \{1, \ldots, n\}$, the probability that there are i agents with a type above t is $\binom{n}{i}(1-t)^i t^{n-i}$, and at most $\min\{i, m\}$ of these agents can be assigned an object. By summing over i, we obtain (on the right-hand side) an upper bound on the expected number of agents with a type above t that can be assigned an object, and the inequality follows.

We have seen in Lemma 1 that whether a mechanism is BIC depends in particular on the lowest probability with which some type obtains an object; for an interim allocation rule P, let therefore $\varphi := \inf_{t'} P(t')$. We then obtain that for any type t,

$$n \int_t^1 P(x)\,\mathrm{d}x \leqslant c^{ic}_{\varphi}(t), \tag{5}$$

where

$$c^{ic}_{\varphi}(t) := m - nt\varphi.$$

Similarly to inequality (4), the left-hand side in (5) denotes the expected number of agents with a type above t receiving an object. Because there are m objects and because in expectation nt agents have a type below t and each of them receives an object with a probability of at least φ, the inequality follows.[9]

Since at most k audits are available, we obtain in analogy with inequality (4) the following feasibility constraint on A that must hold for all types t:

$$n \int_t^1 A(x)\,\mathrm{d}x \leqslant \sum_{i=1}^{n} \min\{i, k\} \binom{n}{i} (1-t)^i t^{n-i}.$$

Lemma 1 implies that if (P, A) is BIC then $A(t) \geqslant P(t) - \inf_{t'} P(t')$. Plugging this into the inequality above and using $\varphi = \inf_{t'} P(t')$, we obtain that for any type t,

$$n \int_t^1 P(x)\,\mathrm{d}x \leqslant c^{aud}_{\varphi}(t), \tag{6}$$

where

$$c^{aud}_{\varphi}(t) := \sum_{i=1}^{n} \min\{i, k\} \binom{n}{i} (1-t)^i t^{n-i} + n(1-t)\varphi.$$

[9]Formally, the inequality follows because (4) implies $n \int_0^t P(x)\,\mathrm{d}x + n \int_t^1 P(x)\,\mathrm{d}x \leqslant m$ and $\varphi \leqslant P(t)$ by definition.

Finally, we combine the constraints in inequalities (4), (5), and (6) to obtain

$$n \int_t^1 P(x)\,\mathrm{d}x \leqslant c_\varphi(t) := \min\{c^{allo}(t), c^{aud}_\varphi(t), c^{ic}_\varphi(t)\}. \tag{7}$$

One can view inequality (7) as being parametrized by φ; any feasible and BIC pair of interim rules (P, A) with $\inf_{t'} P(t') = \varphi$ must satisfy this parametrized inequality.

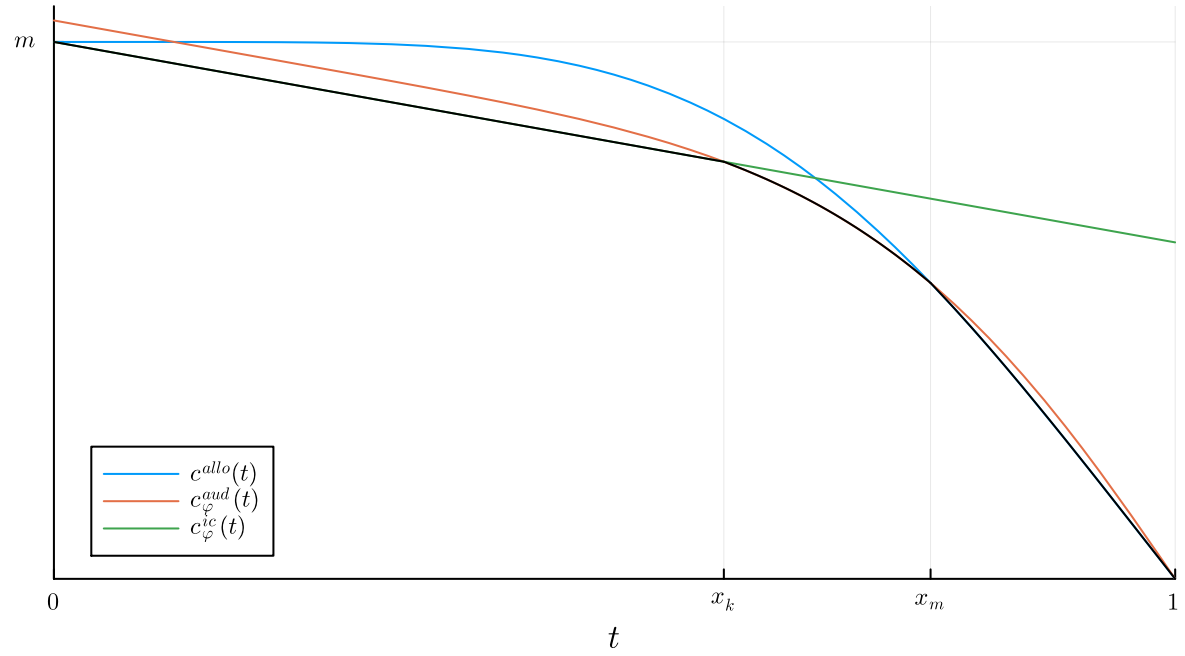


Figure 3: This figure illustrates the right-hand sides of (4)–(6) as functions of t for an example where $n = 8$, $m = 3$, $k = 2$ and $\varphi = 0.14$.

Figure 3 shows the right-hand sides of (4)–(6) for an example where $n = 8$, $m = 3$, $k = 2$ and $\varphi = 0.14$. As can be seen in the figure, constraint (4) is the tightest for high types, constraint (6) for intermediate types, and constraint (5) for low types. Lemma 7 in Appendix A.3 shows that the regions of tightest constraints always take this form: for every $\varphi \in [0, \frac{m}{n}]$, there are unique cutoffs $0 \leqslant x_k(\varphi) \leqslant x_m(\varphi) \leqslant 1$ such that

$$c_\varphi(t) = \begin{cases} c^{ic}_\varphi(t), & 0 < t < x_k(\varphi), \\ c^{aud}_\varphi(t), & x_k(\varphi) < t < x_m(\varphi), \\ c^{allo}(t), & x_m(\varphi) < t < 1. \end{cases} \tag{8}$$

We denote the corresponding regions by $T^\ell = [0, x_k(\varphi)]$, $T^k = (x_k(\varphi), x_m(\varphi)]$, and $T^m = (x_m(\varphi), 1]$, and we suppress the dependence of the cutoffs on φ when it is clear from the context.

These constraints restrict the principal in how often she can allocate to agents with a type above any t. Since the principal aims to allocate as often as possible to high

types, allocation rules that satisfy the tightest constraint with equality for every t are particularly attractive, as formalized in the following result and illustrated in Figure 4.

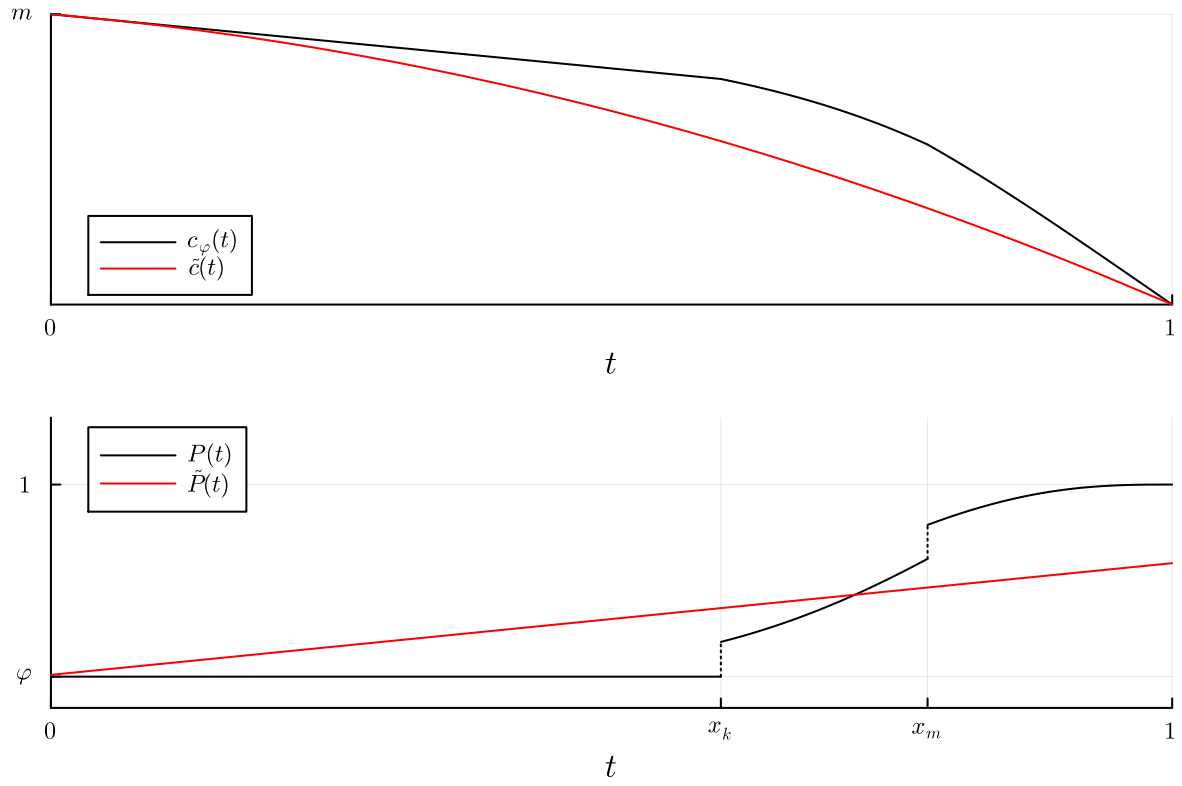


Figure 4: The upper panel illustrates the left- and right-hand sides of (7) for an arbitrary mechanism $\tilde{P}$. The lower panel shows the interim allocation rule $\tilde{P}$ and the hypothetical allocation rule that makes (7) binding, illustrating how $\tilde{P}$ allocates less often to high types.

Lemma 2. *For any feasible and BIC mechanism* $(\tilde{p}, \tilde{a})$ *with* $\varphi = \inf_t \mathbb{E}_{\boldsymbol{t}_{-i},s}[\tilde{p}_i(t, \boldsymbol{t}_{-i}, s)]$, *the principal's payoff is at most*

$$\int_0^1 c_\varphi(t)\,\mathrm{d}v(t).$$

It achieves this upper bound if and only if (7) *holds as equality for all* t.

Proof. Using integration by parts and $v(0) = 0$, we obtain

$$\begin{aligned} n\int_0^1 \tilde{P}(t)\; v(t)\,\mathrm{d}t &= n\; v(t)\int_0^t \tilde{P}(x)\,\mathrm{d}x\Big|_{t=0}^1 - n\int_0^1\int_0^t \tilde{P}(x)\,\mathrm{d}x\,\mathrm{d}v(t) \\ =& n\int_0^1\int_t^1 \tilde{P}(x)\,\mathrm{d}x\,\mathrm{d}v(t) \leqslant \int_0^1 c_\varphi(t)\,\mathrm{d}v(t) \end{aligned}$$

where the last inequality follows from (7). Equality holds if and only if (7) is an equality for all t. □

We now illustrate in the context of the example in Figure 3 why a merit-with-guarantee mechanism with the cutoffs in (8) induces (7) to be binding everywhere (and why only merit-with-guarantee mechanisms can do so). So consider such a merit-with-guarantee mechanism (p, a) with guarantee φ, where $0 < x_k < x_m < 1$ as in the figure, and suppose it satisfies $\mathbb{E}_{\boldsymbol{t}_{-i}}[p_i(t, \boldsymbol{t}_{-i})] \geqslant \varphi$ for all $t \in T$. We verify that this mechanism satisfies (7) as an equality for every t: For any $t \geqslant x_m$, under this allocation rule the expected number of objects allocated to agents with types above t is $\sum_{i=1}^{n} \min\{i, m\}\binom{n}{i}(1-t)^i t^{n-i}$, which equals $c^{allo}(t)$; hence, (7) is an equality for all $t \in T^m$. Also, for any $t \in T^k$, the expected number of objects allocated to agents with types above t is

$$\sum_{i=1}^{n} \min(i, k)\binom{n}{i}(1-t)^i t^{n-i} + \sum_{i=1}^{n}[\min(i, m) - \min(i, k)]\binom{n}{i}(1-x_m)^i x_m^{n-i} + n\varphi(x_m - t);$$

the first sum captures that the k highest agents get an object if their type is above t, the second sum captures that agents with types in T^m also get an object if their types are among the m highest but not among the k highest (these agents are not counted in the first sum), and the last term captures the expected allocation to agents with types above t in the lottery stage. Using the definitions of c^{allo} and c_φ^{aud}, we can simplify the last display to

$$c_\varphi^{aud}(t) + c^{allo}(x_m) - c_\varphi^{aud}(x_m) = c_\varphi^{aud}(t),$$

where the last equality uses that, by (8) and continuity, $c^{allo}(x_m) = c_\varphi^{aud}(x_m)$. Hence, for any $t \in [x_k, x_m)$, (7) holds as an equality. Finally, let $t \in [0, x_k]$. Agents with types in $[t, x_k]$ obtain an object only in the lottery, with probability φ. By the previous step at $t = x_k$ and since $c_\varphi^{aud}(x_k) = c_\varphi^{ic}(x_k)$ by (8), the expected number of objects allocated to agents with types above t is

$$c_\varphi^{aud}(x_k) + n\varphi(x_k - t) = m - n\varphi x_k + n\varphi(x_k - t) = c_\varphi^{ic}(t).$$

In particular, for $t = 0$, the expected number of allocated objects is $c_\varphi^{ic}(0) = m$. Since there are only m objects, such a mechanism allocates all objects almost surely. Jointly, these observations imply that if the merit-with-guarantee mechanism illustrated in Figure 3 satisfies $\mathbb{E}_{\boldsymbol{t}_{-i}}[p_i(t, \boldsymbol{t}_{-i})] \geqslant \varphi$ for all $t \in T$, then it attains the upper bound $\int_0^1 c_\varphi(t)\,\mathrm{d}v(t)$ in Lemma 2, which is the highest payoff among all mechanisms with guarantee φ; moreover, any mechanism with guarantee φ that attains this bound must take this form almost everywhere.

For the general proof, we consider for a given $\varphi \in [\frac{m-k}{n}, \frac{m}{n}]$ (smaller guarantees yield a lower upper bound, see Appendix A.6) the intervals T^m, T^ℓ, and T^k defined above and the merit-based allocation rule p^M with thresholds x_k and x_m. We show in Lemma 9 that we can complement this allocation rule by a lottery-based rule for the second stage so that each type has a probability at least φ of getting an object under the combined rule

p; moreover, (7) will be binding for all t. Lemma 10 implies that there exists an audit rule a that is feasible given p such that (p, a) is incentive compatible. By Lemma 2, (p, a) attains the upper bound $\int_0^1 c_\varphi(t)\,\mathrm{d}v(t)$, and choosing φ to maximize this bound yields an optimal mechanism.

5 Optimal guarantee and large market analysis

5.1 Optimal guarantee

In this section, we discuss how to optimally choose the thresholds x_k and x_m for the merit-with-guarantee mechanism. The proof of Theorem 1 reveals that finding the optimal thresholds can be reduced to optimizing the guarantee φ, a single parameter, and then using the cutoffs $x_k(\varphi)$ and $x_m(\varphi)$ defined in (8).

Which guarantees φ can be optimal? It turns out that for optimal values of φ, all three regions are nondegenerate and we can describe optimal values using a simple first-order condition.

Proposition 1. *Assume v is continuous. Every optimal guarantee φ satisfies $\frac{m-k}{n} < \varphi < \frac{m}{n}$ and $0 < x_k(\varphi) < x_m(\varphi) < 1$, so that T^ℓ, T^k, and T^m are nondegenerate intervals. Moreover, $\varphi \in (\frac{m-k}{n}, \frac{m}{n})$ is optimal if and only if*

$$v(x_k(\varphi)) = (1 - x_m(\varphi))\, v(x_m(\varphi)) + \int_0^{x_m(\varphi)} v(t)\,\mathrm{d}t. \tag{9}$$

The first-order condition balances two effects of a marginal increase in φ. On the one hand, a higher guarantee tightens the incentive constraint (5): more objects must be allocated to low types, which lowers $c^{ic}_\varphi(t)$ at rate nt for types $t \in T^\ell$. On the other hand, it relaxes the audit constraint (6): fewer audits are needed to deter misreports, which raises $c^{aud}_\varphi(t)$ at rate $n(1-t)$ for types $t \in T^k$. The allocation constraint, which binds on T^m, is unaffected. Since the principal's payoff is $\int_0^1 c_\varphi(t)\,\mathrm{d}v(t)$, an optimal guarantee balances these two effects,

$$\int_0^{x_k} t\,\mathrm{d}v(t) = \int_{x_k}^{x_m} (1-t)\,\mathrm{d}v(t),$$

and integration by parts yields (9).

We briefly explain why all three regions are used in an optimal mechanism. To see that T^m is nonempty, note that because there are only m objects, φ can be at most $\frac{m}{n}$. Moreover, $\varphi = \frac{m}{n}$ corresponds to allocating uniformly at random, which is dominated by slightly smaller values of φ. Indeed, whenever $0 < \varphi < \frac{m}{n}$, c^{allo} is strictly below c^{ic}_φ and c^{aud}_φ in a neighborhood around $t = 1$, and T^m is nonempty and allocations are efficient for the highest types.

To see that T^ℓ is nonempty, suppose to the contrary that it is empty. This implies $c^{ic}_\varphi(0) \geqslant c^{aud}_\varphi(0)$. A slightly higher value of φ increases c^{aud}_φ pointwise, which expands T^m (see Figure 3) and increases allocation probabilities for higher types at the cost of types close to 0, improving the principal's payoff and contradicting optimality.

Finally, to see that T^k is nonempty, suppose φ is high enough that c^{aud}_φ lies everywhere above either c^{ic}_φ or c^{allo}, so that T^k is empty. Then reducing φ lowers c^{aud}_φ pointwise, expanding T^m, since the intersection between c^{ic}_φ and c^{allo} shifts to the left, and increasing allocation probabilities for higher types at the cost of lower types. As φ decreases further, T^k eventually consists of a single point. Beyond this, reducing φ introduces a tradeoff: T^m shrinks, so efficient allocation occurs for fewer types, but T^ℓ also shrinks, so allocation becomes more efficient for intermediate types. The proof of Proposition 1 shows via the first-order condition that small reductions in φ remain beneficial, implying that T^k is nondegenerate at the optimum.

Example 2. *Suppose that there are three agents, two objects to allocate, and only one agent can be verified: $n = 3, m = 2, k = 1$. We assume $v(t) = t$ for all t.*

We now illustrate how to use Proposition 1 to find an optimal value of φ. This corresponds to determining which constraint function is binding for the upper bound,

$$3\int_t^1 P(x)\,\mathrm{d}x \leqslant c_\varphi(t) = \min\{c^{allo}(t), c^{aud}_\varphi(t), c^{ic}_\varphi(t)\}.$$

For this example the constraint functions are the following:

$$c^{allo}(t) = t^3 - 3t^2 + 2,\ c^{aud}_\varphi(t) = -t^3 - 3\varphi t + 1 + 3\varphi,\ \text{and } c^{ic}_\varphi(t) = 2 - 3\varphi t,$$

where the optimal value of φ lies in $[\frac{1}{3}, \frac{2}{3}]$. The interim allocation rule of the merit-with-guarantee mechanism with guarantee φ is

$$P(t) = \begin{cases} \varphi & \text{for } t \in [0, x_k(\varphi)] \\ t^2 + \varphi & \text{for } t \in (x_k(\varphi), x_m(\varphi)] \\ 2t - t^2 & \text{for } t \in (x_m(\varphi), 1]. \end{cases}$$

The expected per-agent payoff for the principal as a function of φ is

$$U(\varphi) = \int_0^{x_k(\varphi)} \varphi t\,\mathrm{d}t + \int_{x_k(\varphi)}^{x_m(\varphi)} t^3 + \varphi t\,\mathrm{d}t + \int_{x_m(\varphi)}^1 2t^2 - t^3\,\mathrm{d}t.$$

Using Proposition 1 we have the following condition for φ to be optimal:

$$x_k(\varphi) = x_m(\varphi) - \frac{x_m(\varphi)^2}{2}.$$

Solving this equation gives us the optimal value $\varphi^* \approx 0.348$ and an expected total payoff of 1.223. This can be compared with the expected payoff of 1 from using a random allocation

rule where all agents get an object with an equal probability of $\frac{2}{3}$*. Another comparison is with the first-best mechanism that always allocates the two objects to the agents with the highest types, yielding an expected payoff of* 1.25*.*

We now illustrate one construction of a lottery rule p^ℓ *for this example. At the lottery stage, all agents with reports in* T^ℓ *are treated symmetrically: their winning probability depends only on the number of lottery participants, the number of available objects, and whether any participant reports a type in* T^k*. For lottery instances involving only agents from* T^ℓ*, if two agents compete for one object, each wins with probability* $1/2$*; if three such agents compete for two objects, each wins with probability* $2/3$*. When one agent reports from* T^ℓ *and another from* T^k*, with one object available, we set the winning probabilities to* 0.6872 *for the* T^k *agent and* 0.3128 *for the* T^ℓ *agent. This ensures that all types in* T^ℓ *receive an object in the lottery stage with interim probability* φ^**.*

To complete the description of the lottery rule, we must specify winning probabilities when both agents report types in T^k*. Since agents in* T^k *enter the lottery only when they fail to receive an object in the merit stage, the probability of entering the lottery decreases with their report (higher reports are more likely to rank among the top* k*). Therefore, to ensure all reports in* T^k *obtain an object at the lottery stage with interim probability* φ^**, the conditional winning probability in the lottery must increase in their report. When two agents with reports* $t, t' \in T^k$ *both enter the lottery, we set the winning probability for the agent with report* t *to* $r(t,t') = \frac{1}{2} + g(t) - g(t')$ *for some strictly increasing function* g*. This ensures that the object is always assigned:* $r(t,t') + r(t',t) = 1$*. The function* g *is determined, up to an additive constant, by requiring that each agent with report* $t \in T^k$ *receives an object at the lottery stage with interim probability* φ^**.*

5.2 Large market analysis

While typical applications we have in mind involve a moderate number of agents and objects, in this section we analyze large markets to better understand limiting properties of the optimal mechanisms. Specifically, consider a sequence of markets indexed by n, where market n has n agents, m_n objects, and audit capacity k_n, with $m_n/n \to \alpha$ and $k_n/n \to \beta$ as $n \to \infty$ for some fixed $0 < \beta < \alpha < 1$. Let P_n denote the interim allocation rule of an optimal merit-with-guarantee mechanism in a market of size n, let φ_n denote its guarantee, and let $A_n := P_n - \varphi_n$ denote the corresponding interim audit rule that audits as rarely as incentive compatibility permits.

Proposition 2. *As* $n \to \infty$*, the optimal guarantee satisfies* $\varphi_n \to \alpha - \beta$*, the optimal interim allocation rule satisfies*

$$P_n(t) \to \begin{cases} \alpha - \beta & \text{if } t < \frac{1-\alpha}{1-\alpha+\beta} \\ 1 & \text{if } t > \frac{1-\alpha}{1-\alpha+\beta}, \end{cases}$$

and the interim audit rule satisfies

$$A_n(t) \to \begin{cases} 0 & \text{if } t < \frac{1-\alpha}{1-\alpha+\beta} \\ 1-(\alpha-\beta) & \text{if } t > \frac{1-\alpha}{1-\alpha+\beta}. \end{cases}$$

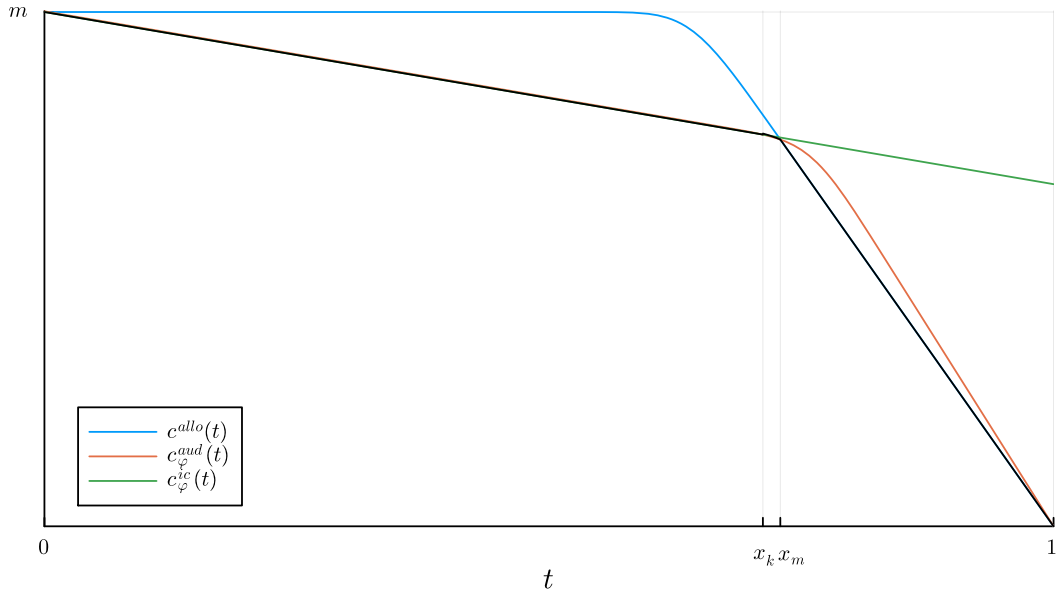


Figure 5: This figure illustrates the right-hand sides of (4)–(6) as functions of t for $n = 250$, $m = 90$, $k = 60$, and $\varphi = 0.1205$.

Intuitively, since an optimal mechanism allocates all objects, Lemma 1 implies that the expected number of audits is at least $m_n - n\varphi_n$, and since at most k_n audits are available, $\varphi_n \geqslant \frac{m_n - k_n}{n}$. In a finite market, the number of reports in T^m is random. When fewer than k_n agents report in T^m, some audits would remain idle, and the intermediate region T^k uses them: reports in T^k that rank among the k_n highest obtain an object on merit and are audited. In large markets, the share of reports in T^m concentrates around a value strictly between the share β of audits and the share α of objects. With probability approaching one, every report in T^m then obtains an object and no report in T^k ranks among the k_n highest, so that essentially all audit capacity can be used. In the limit, a guarantee above $\alpha - \beta$ would require fewer audits than the available capacity; lowering the guarantee would allow the principal to expand T^m, shifting objects toward higher types. Hence, the optimal guarantee converges to its lower bound $\alpha - \beta$, and the limit cutoff is determined by the requirement that all objects are allocated when every type above the cutoff obtains an object and every type below it obtains one with probability $\alpha - \beta$. Figure 5 illustrates the corresponding constraint functions for a large market.

Proposition 2 does not imply that the interval T^k vanishes in large markets. Its upper end x_m converges to the limit cutoff (see the proof in Appendix B.2), but its lower end x_k

need not: for the limiting guarantee $\alpha - \beta$, the audit and incentive constraints coincide in the limit for all types below the cutoff, so they do not pin down x_k. Instead, x_k is determined by the first-order condition (9); for example, if $v(t) = t$, the length of T^k converges to $\frac{1}{2}\left(\frac{1-\alpha}{1-\alpha+\beta}\right)^2 > 0$. What vanishes is the role of T^k in merit-based allocation: as explained above, an agent with a type in T^k almost never ranks among the k_n highest in a large market, so he obtains an object essentially only in the lottery, and his allocation probability is close to the guarantee.

Corollary 1. *As α increases, the limit cutoff decreases and the allocation probability for types below this cutoff increases.*

As β increases, the limit cutoff decreases and the allocation probability for types below this cutoff decreases.

As α increases, more objects become available, which allows the principal to decrease the cutoff above which the allocation is efficient. Since this alone would make misreports more attractive, the probability for low reports to receive an object in the lottery increases. As β increases, more audits become available. This allows the principal to reduce the probability for low reports to get an object and thereby to reduce the cutoff for efficient allocation.

Based on Proposition 2, we can describe a simple mechanism that, for all sufficiently large markets, is incentive compatible and achieves a per-capita payoff close to that of the optimal mechanism. Let $r_n := \lfloor nk_n/(n - m_n + k_n) \rfloor$ and call a report high if it exceeds $1 - r_n/n$. In the first round, all agents with high reports are selected if there are at most r_n of them; otherwise, r_n of them are selected uniformly at random. Among the selected agents, k_n are audited, chosen uniformly at random (all of them if fewer are selected), and every selected agent obtains an object unless his report is found to be false. In the second round, the remaining objects are allocated uniformly at random among agents with low reports; if fewer low reports than remaining objects are available, each agent receives one object and the excess remains unallocated. As n grows, the cutoff $1 - r_n/n$ converges to the limit cutoff $\frac{1-\alpha}{1-\alpha+\beta}$. Lemma 11 in Appendix B.2 shows that this mechanism is Bayesian incentive compatible for all sufficiently large n and that its per-capita payoff converges to the limit of the optimal per-capita payoff.

This mechanism is simple in that it uses only a single cutoff and in that it selects agents uniformly at random. As the market size grows, the probability of a type above the cutoff receiving an object converges to 1, and the probability of an audit conditional on being selected for allocation converges to $1 - \alpha + \beta$. So in this mechanism, agents have similar allocation probabilities as in the optimal merit-with-guarantee mechanism in large markets. The merit-with-guarantee rule dominates this simple mechanism when there are many agents with high types because it always allocates efficiently within the highest

interval. In contrast, the simple mechanism randomly rations high types, allocating at most r_n objects to them and using a lottery to allocate the remaining objects among agents reporting below the cutoff. If there are fewer than $m_n - r_n$ low reports, some objects remain unallocated.

6 Conclusion

We have analyzed the problem of allocating m identical objects to n agents when only k checks are available to verify agents' private information and monetary transfers are unavailable, and have shown that a merit-with-guarantee mechanism is optimal. A key insight is that guaranteeing low types a positive allocation probability φ sustains truthful reporting and in turn allows the mechanism to allocate efficiently beyond the k agents that can be verified.

To illustrate, consider the social housing application discussed in the introduction. A merit-with-guarantee mechanism partitions applicants into three groups based on reported housing need. Those reporting the highest need fall in the region T^m, and are allocated housing in the first stage whenever their reports are among the m highest, even though not all of them can be verified. Applicants reporting moderately high need, in the region T^k, are also allocated housing in the first stage but only when their reports are among the k highest, in which case they are verified with certainty and lose their allocation if found to have misrepresented their circumstances. Applicants reporting lower levels of need, in the region T^ℓ, enter the second-stage lottery together with applicants from T^k whose reports are not among the k highest; all of them receive housing with the guaranteed probability.

Constraints on verification arise naturally in many allocation settings beyond social housing. In the allocation of research grants, for instance, funding bodies rely on peer review to assess proposal quality, but reviewer capacity limits the number of proposals that can be scrutinized in depth. Our results suggest that in such settings, complementing merit-based allocation with a guarantee—a positive allocation probability for agents whose reports are not verified—can improve outcomes by sustaining truthful reporting while allowing the mechanism to allocate efficiently beyond its verification capacity. While we are not aware of procedures that explicitly advertise such guarantees, they can be interpreted as an implicit feature of systems that combine merit-based assignment with truncated reporting, where applicants below a threshold are pooled and treated symmetrically.

A Proof of Theorem 1

In the remainder of the proof of Theorem 1, we will show that, for any $\varphi \in [\frac{m-k}{n}, \frac{m}{n}]$, the merit-based allocation rule determined by c_φ can be complemented by a lottery rule p^ℓ satisfying $\mathbb{E}_{\boldsymbol{t}_{-i},s}[p_i^\ell(t_i, \boldsymbol{t}_{-i}, s)] = \varphi$ for all $t_i \in T^\ell \cup T^k$ and by an audit rule a satisfying $\mathbb{E}_{\boldsymbol{t}_{-i},s}[a_i(t_i, \boldsymbol{t}_{-i}, s)] \geqslant \mathbb{E}_{\boldsymbol{t}_{-i},s}[p_i(t_i, \boldsymbol{t}_{-i}, s)] - \varphi$ for all i and all $t_i \in T$.[10] We also show that the induced interim allocation rule is equal to $-\frac{1}{n}c'_\varphi$. By Lemma 2, the resulting mechanism therefore attains the upper bound $\int_0^1 c_\varphi(t)\,\mathrm{d}v(t)$, and choosing φ to maximize this bound yields an optimal mechanism.

We say that the partition $\{T^\ell, T^k, T^m\}$ is *determined by* c_φ if $T^\ell = [0, x_k]$, $T^k = (x_k, x_m]$, and $T^m = (x_m, 1]$, where $x_k = x_k(\varphi)$ and $x_m = x_m(\varphi)$ are the cutoffs in (8). By Lemma 7, c_φ then equals c_φ^{ic}, c_φ^{aud}, and c^{allo} on T^ℓ, T^k, and T^m, respectively.

A.1 Characterization of BIC mechanisms

Proof of Lemma 1. Let $i \in N$. By the definition of BIC in (1) and the fact that $u_i(t_i) > 0$ for all $t_i \in T$ we get that agent i with type $t_i \in T$ has no incentive to deviate if and only if, for all $t'_i \in T$,

$$\mathbb{E}_{\boldsymbol{t}_{-i},s}[p_i(t_i, \boldsymbol{t}_{-i}, s)] \geqslant \mathbb{E}_{\boldsymbol{t}_{-i},s}[p_i(t'_i, \boldsymbol{t}_{-i}, s)[1 - a_i(t'_i, \boldsymbol{t}_{-i}, s)]]. \tag{10}$$

Since $a_i(\boldsymbol{t}, s) = 1$ only if $p_i(\boldsymbol{t}, s) = 1$, the right-hand side equals $P(t'_i) - A(t'_i)$. Since (10) is required to hold for all $t_i \in T$, it must in particular hold for the infimum over T. Thus, Definition 1 of BIC is equivalent to the inequality in Lemma 1. □

A.2 Feasibility of interim rules

We consider the interim rules induced by the merit-based rule p^M and lottery rule p^ℓ and the audit rule a and show that they satisfy the necessary feasibility constraints for interim rules known from the literature.

To begin with, we introduce some notation and definitions that we will use throughout the appendix. For $D_1, D_2 \subseteq [0,1]$, the disjoint union is defined as $D_1 \sqcup D_2 := \{(t,1) | t \in D_1\} \cup \{(t,2) | t \in D_2\}$, and analogously for disjoint unions of n sets. In contrast to the standard union, the disjoint union indexes each element by the set it comes from. For every $E = E_1 \sqcup \cdots \sqcup E_n$ with each $E_i \subseteq T$, and every $\boldsymbol{t} \in \boldsymbol{T}$, let $I(\boldsymbol{t}, E) := \{i \in N | t_i \in E_i\}$ denote the set of agents with types in E that are compatible with $\boldsymbol{t}$. Slightly abusing notation, if $E \subseteq T$, we write $I(\boldsymbol{t}, E)$ for $I(\boldsymbol{t}, E \sqcup \ldots \sqcup E)$.

[10] It suffices to consider guarantees $\varphi \geqslant \frac{m-k}{n}$, since smaller guarantees cannot be optimal (see Section A.6).

In Section 4 we argued that the interim allocation rule of any feasible BIC mechanism must respect the upper bounds c^{allo} and c^{aud}_{φ}, due to the limited availability of objects and audits. Formally, this can be derived from the following result, which extends the characterization of feasible interim rules by Border (1991) from one object to m objects.

Lemma 3. *An interim allocation rule P is feasible if and only if for all Borel sets $E \subseteq T$,*

$$n \int_E P(t_i)\,\mathrm{d}t_i \leqslant \int \min\{|I(\boldsymbol{t}, E)|, m\}\,\mathrm{d}\boldsymbol{t}. \tag{11}$$

Moreover, if P is non-decreasing, it is enough to check this inequality for sets E of the form $E = [e, 1]$.

The next lemma, which follows from Theorem 6 in Valenzuela-Stookey (2024), characterizes interim rules in our setting with type-dependent ex-post constraints. Specifically, we assume that if type profile $\boldsymbol{t}$ realizes, $h(\boldsymbol{t})$ objects can be allocated. Moreover, we assume only agents in the set $J(\boldsymbol{t})$ can get an object. An allocation rule p is *feasible* if, for all $\boldsymbol{t} \in \boldsymbol{T}$ and $s \in [0,1]$, $\sum_i p_i(\boldsymbol{t}, s) \leqslant h(\boldsymbol{t})$ and $p_i(\boldsymbol{t}, s) = 1$ implies $i \in J(\boldsymbol{t})$.[11] To formulate the lemma we consider asymmetric interim rules $P_i : T \to [0,1]$ where $P_i(t) := \mathbb{E}_{\boldsymbol{t}_{-i},s}[p_i(t, \boldsymbol{t}_{-i}, s)]$.

Lemma 4. *Let $h(\boldsymbol{t})$ be the total number of objects available, and let $J(\boldsymbol{t})$ denote the set of agents that can receive an object, which can both depend on the realized type profile. There is a feasible allocation rule $p = (p_i)_{i \in N}$ inducing an interim allocation rule $P = (P_i)_{i \in N}$ if and only if, for all sets $E = \bigsqcup_{i \in N} E_i$, where each E_i is a Borel subset of T,*

$$\sum_i \int_{E_i} P_i(t)\,\mathrm{d}t \leqslant \int \min\{|J(\boldsymbol{t}) \cap I(\boldsymbol{t}, E)|, h(\boldsymbol{t})\}\,\mathrm{d}\boldsymbol{t}.$$

If, moreover, $P_1 = \cdots = P_n$ and h and J are symmetric,[12] it is sufficient to check this inequality for sets with $E_1 = \cdots = E_n$.

Note that for an interim audit rule A to be feasible there must exist an audit rule a with $A(t) = \mathbb{E}_{\boldsymbol{t}_{-i},s}[a_i(t, \boldsymbol{t}_{-i}, s)]$ for each $t \in T$, and $a(\boldsymbol{t}, s) \leqslant p(\boldsymbol{t}, s)$, for the specific p that induces P. We imposed the latter constraint $a(\boldsymbol{t}, s) \leqslant p(\boldsymbol{t}, s)$, since this simplified our characterization of incentive compatibility. However, this additional constraint complicates the characterization of which interim audit rules are feasible given an allocation rule.

To state the characterization, let p be a symmetric allocation rule that is deterministic for almost all $\boldsymbol{t}$. Define for every $E = E_1 \sqcup \cdots \sqcup E_n$ with each $E_i \subseteq T$, and every $\boldsymbol{t} \in \boldsymbol{T}$, $J^p(\boldsymbol{t}, E) := \{i \in N | t_i \in E_i \text{ and } p_i(\boldsymbol{t}) = 1\}$. Slightly abusing notation, if $E \subseteq T$, we write $J^p(\boldsymbol{t}, E)$ for $J^p(\boldsymbol{t}, E \sqcup \ldots \sqcup E)$.

[11] With slight abuse of notation, we will drop the realization of the randomization device as an argument for rules that are deterministic for almost every $\boldsymbol{t}$.

[12] That is, for all $\boldsymbol{t}$ and permutations σ, $h(\boldsymbol{t}) = h(\boldsymbol{t}_\sigma)$, and $i \in J(\boldsymbol{t})$ if and only if $\sigma(i) \in J(\boldsymbol{t}_\sigma)$.

Lemma 5. *An interim audit rule A is feasible given a symmetric and deterministic ex-post allocation rule p if and only if for all Borel sets $E \subseteq T$,*

$$n \int_E A(t_i)\,\mathrm{d}t_i \leqslant \int \min\{|J^p(\boldsymbol{t}, E)|, k\}\,\mathrm{d}\boldsymbol{t}. \tag{12}$$

Although the interim rules induced by p^ℓ and a are monotone, it does not suffice to check feasibility constraints in Lemma 4 and 5 only for upper sets. This is because the number of objects available to allocate (or the number of audits available) is effectively dependent on the type profile: the number of objects available in the lottery stage depends on how many objects are allocated in the merit stage, and therefore on the type profile; similarly, because our formulation of incentive compatibility assumes that only agents receiving an object can be audited, the number of audits that can be conducted depends on the type profile. If the number of objects is type-dependent, it is not sufficient to check the Border constraints only for upper sets.[13]

A.3 Properties of the function c_φ and the partition of T determined by c_φ

In this subsection we derive properties of the functions c^{allo}, c_φ^{aud}, and c_φ^{ic}; we will use these properties later to show that there is a merit-with-guarantee mechanism for which (7) is binding for all t.

Lemma 6. *The derivatives of c^{allo} and c_φ^{aud} are given by*

$$\frac{d}{dt}(c^{allo}(t)) = -n \sum_{j=0}^{m-1} \binom{n-1}{j} t^{n-1-j}(1-t)^j \tag{13}$$

$$\frac{d}{dt}(c_\varphi^{aud}(t)) = -n \sum_{j=0}^{k-1} \binom{n-1}{j} t^{n-1-j}(1-t)^j - n\varphi. \tag{14}$$

Also, $c^{allo}(0) = c_\varphi^{ic}(0) = m$, $c_\varphi^{aud}(0) = k + n\varphi$, and $c^{allo}(1) = c_\varphi^{aud}(1) = 0$. Moreover, c^{allo}

[13] Indeed, consider an example with two agents and two types each, $T = \{\underline{t}, \bar{t}\}$ with $\underline{t} < \bar{t}$. Types are iid and both types are equally likely. Further, all agents can be assigned at most one object at all type profiles but the number of objects available at a given type profile varies. The following table specifies how many objects are available:

	$\underline{t}_2$	$\bar{t}_2$
$\underline{t}_1$	0	1
$\bar{t}_1$	2	0

Now, consider the following monotone interim rule $P_1(\underline{t}_1) = P_1(\bar{t}_1) = P_2(\underline{t}_2) = 0.25$ and $P_2(\bar{t}_2) = 0.5$. One can verify that the Border constraints are satisfied for all upper sets, but violated for the set $\{\underline{t}_1, \bar{t}_2\}$. Thus, the interim rule is not feasible.

and c_φ^{aud} are strictly concave, with

$$(c^{allo})''(t) = -nm\binom{n-1}{m}t^{n-m-1}(1-t)^{m-1},$$
$$(c_\varphi^{aud})''(t) = -nk\binom{n-1}{k}t^{n-k-1}(1-t)^{k-1}.$$

Proof of Lemma 6. Recall that $c^{allo}(t)$ is the expected number of agents with types above t who receive an object when the m objects are allocated to the agents with the highest types. Equivalently, it is the expected minimum of m and the number of agents with types above t. At every type profile at which no two types coincide, this minimum equals the number of agents whose types exceed t and are among the m highest types. An agent with type s is among the m highest types if and only if at most $m-1$ of the other $n-1$ agents have types above s, which happens with probability $\sum_{j=0}^{m-1}\binom{n-1}{j}s^{n-1-j}(1-s)^j$. Hence,

$$c^{allo}(t) = n\int_t^1 \sum_{j=0}^{m-1}\binom{n-1}{j}s^{n-1-j}(1-s)^j\,\mathrm{d}s,$$

and differentiating yields (13). The same argument with k in place of m, together with $\frac{d}{dt}\big(n\varphi(1-t)\big) = -n\varphi$, yields (14). The values at $t=0$ and $t=1$ follow directly from the definitions.

To compute the second derivatives, we use $(n-1-j)\binom{n-1}{j} = (n-1)\binom{n-2}{j}$ and $j\binom{n-1}{j} = (n-1)\binom{n-2}{j-1}$. Then the derivative of the sum in (13) telescopes (with $\binom{n-2}{-1}=0$):

$$\begin{aligned}
&\frac{d}{dt}\sum_{j=0}^{m-1}\binom{n-1}{j}t^{n-1-j}(1-t)^j \\
&\quad= (n-1)\sum_{j=0}^{m-1}\left[\binom{n-2}{j}t^{n-2-j}(1-t)^j - \binom{n-2}{j-1}t^{n-1-j}(1-t)^{j-1}\right] \\
&\quad= (n-1)\binom{n-2}{m-1}t^{n-m-1}(1-t)^{m-1}.
\end{aligned}$$

Since $(n-1)\binom{n-2}{m-1} = m\binom{n-1}{m}$, this gives the formula for $(c^{allo})''$, and the same computation with k in place of m gives the formula for $(c_\varphi^{aud})''$. Both second derivatives are strictly negative on $(0,1)$, so c^{allo} and c_φ^{aud} are strictly concave. □

The next lemma shows that the three constraints are tightest on three consecutive intervals.

Lemma 7. *For every $\varphi \in [0, \frac{m}{n}]$, there exist unique $0 \leqslant x_k \leqslant x_m \leqslant 1$ such that*

$$c_\varphi(t) = \begin{cases} c_\varphi^{ic}(t), & 0 < t < x_k, \\ c_\varphi^{aud}(t), & x_k < t < x_m, \\ c^{allo}(t), & x_m < t < 1. \end{cases}$$

Proof. We first show that the allocation constraint binds on an upper interval. We then use concavity and monotonicity to establish that the incentive constraint binds before the audit constraint, yielding the three-interval structure.

Let c_0^{aud} denote c_φ^{aud} for $\varphi = 0$ and, for $t < 1$, define

$$h(t) := \frac{c^{allo}(t) - c_0^{aud}(t)}{1-t},$$

so that $c^{allo}(t) \leqslant c_\varphi^{aud}(t)$ if and only if $h(t) \leqslant n\varphi$. By Lemma 6,

$$(c^{allo})''(t) = -nm\binom{n-1}{m}t^{n-m-1}(1-t)^{m-1},$$
$$(c_0^{aud})''(t) = -nk\binom{n-1}{k}t^{n-k-1}(1-t)^{k-1}.$$

Differentiating $(1-t)h(t) = c^{allo}(t) - c_0^{aud}(t)$ yields

$$(1-t)^2 h'(t) = (1-t)(c^{allo} - c_0^{aud})'(t) + (c^{allo} - c_0^{aud})(t), \tag{15}$$

and differentiating once more gives

$$\begin{aligned}\frac{d}{dt}\big[(1-t)^2 h'(t)\big] &= (1-t)\big[(c^{allo})''(t) - (c_0^{aud})''(t)\big] \\ &= -nt^{n-m-1}(1-t)^k\left[m\binom{n-1}{m}(1-t)^{m-k} - k\binom{n-1}{k}t^{m-k}\right].\end{aligned}$$

The term in brackets is strictly decreasing from a positive to a negative value, so $(1-t)^2 h'(t)$ first decreases and then increases. By (15) and Lemma 6, $(1-t)^2 h'(t)$ equals $m - k > 0$ at 0 and converges to zero as $t \to 1$. It therefore has a unique zero $z \in (0,1)$, which lies on its decreasing part. Hence h is strictly increasing on $[0,z]$ and strictly decreasing on $[z,1)$, and

$$(c^{allo})''(z) < (c_0^{aud})''(z) < 0.$$

We next show that $c^{allo}(z) > m - (m-k)z$. Using $h'(z) = 0$ and the endpoint values in Lemma 6, one can use integration by parts to show[14]

$$c^{allo}(z) - m + (m-k)z = \int_0^z \Big[z(1-t)(c_0^{aud})''(t) - (1-z)t(c^{allo})''(t)\Big]\,\mathrm{d}t.$$

It is therefore sufficient to show that the integrand is strictly positive for $0 < t < z$. Dividing the formulas for the second derivatives above gives

$$\frac{t(c^{allo})''(t)}{(1-t)(c_0^{aud})''(t)} = \frac{m\binom{n-1}{m}}{k\binom{n-1}{k}}\left(\frac{1-t}{t}\right)^{m-k-1}.$$

[14]Indeed, integrating by parts, $\int_0^z t(c^{allo})''(t)\,\mathrm{d}t = z(c^{allo})'(z) - c^{allo}(z) + m$ and $\int_0^z (1-t)(c_0^{aud})''(t)\,\mathrm{d}t = (1-z)(c_0^{aud})'(z) + c_0^{aud}(z) - k$, and $h'(z) = 0$ implies $(1-z)(c^{allo} - c_0^{aud})'(z) = c_0^{aud}(z) - c^{allo}(z)$.

This ratio is nonincreasing in t, and at $t = z$ it strictly exceeds $z/(1-z)$, because $(c^{allo})''(z)/(c_0^{aud})''(z) > 1$ by the previous inequality. Hence, the ratio exceeds $z/(1-z)$ for all $0 < t \leqslant z$, and the integrand is therefore strictly positive. By concavity of c^{allo} and $c^{allo}(0) = m$, we also obtain

$$c^{allo}(t) > m - (m-k)t \qquad \text{for } 0 < t \leqslant z.$$

The allocation constraint cannot bind on $(0, z]$. Indeed, recall that $c^{allo}(t) \leqslant c_\varphi^{aud}(t)$ if and only if $h(t) \leqslant n\varphi$; hence, if $c^{allo}(t) \leqslant c_\varphi^{aud}(t)$ for some $t \in (0, z]$, then $n\varphi \geqslant h(t) > h(0) = m - k$, which implies

$$c_\varphi^{ic}(t) = m - n\varphi t < m - (m-k)t < c^{allo}(t).$$

On $[z, 1)$, by contrast, $c^{allo} \leqslant c_\varphi^{aud}$ defines an upper interval because h is decreasing. The same is true of $c^{allo} \leqslant c_\varphi^{ic}$, because $c^{allo} - c_\varphi^{ic}$ is strictly concave, vanishes at 0, and is nonpositive at 1. Their intersection is therefore an upper interval, with lower boundary x_m.

Below x_m, the binding constraint is either c_φ^{ic} or c_φ^{aud}. Since $c_\varphi^{aud} - c_\varphi^{ic}$ is strictly decreasing by (14), the incentive constraint binds below a cutoff $x_k \leqslant x_m$, and the audit constraint binds between x_k and x_m.

Finally, the cutoffs are unique because no two of the functions c^{allo}, c_φ^{aud}, and c_φ^{ic} coincide on an interval. □

A.4 Existence of lottery rule p^ℓ

In this subsection of the appendix we show that there exists a feasible lottery rule p^ℓ with $\mathbb{E}_{\boldsymbol{t}_{-i},s}[p_i^\ell(t_i, \mathbf{t}_{-i}, s)] = \varphi$ for all $t_i \in T^\ell \cup T^k$ and that the allocation rule in the merit-with-guarantee mechanism induces the interim rule $-\frac{1}{n}c'_\varphi$.

Fix $\varphi \in [\frac{m-k}{n}, \frac{m}{n}]$ and let the partition $\{T^\ell, T^k, T^m\}$ be determined by c_φ, so that $T^\ell = [0, x_k]$, $T^k = (x_k, x_m]$, and $T^m = (x_m, 1]$. Let p^M denote the corresponding merit-based allocation rule in Definition 2.

Lemma 8. *The interim rule of p^M is $-\frac{1}{n}c'_\varphi - \varphi \mathbf{1}_{T^k \cup T^\ell}$.*

Proof. Note that for type $t_i \in T^m$, the probability that i obtains an object under allocation rule p^M equals the probability that there are at most $m-1$ others with types above t_i. Using $c_\varphi = c^{allo}$ on T^m and Lemma 6, this yields

$$\mathbb{E}_{\boldsymbol{t}_{-i}}[p_i^M(t_i, \boldsymbol{t}_{-i})] = \sum_{j=0}^{m-1} \binom{n-1}{j} t_i^{n-1-j}(1-t_i)^j = -\frac{1}{n}c'_\varphi(t_i).$$

If $t_i \in T^k$, the probability that i obtains an object under allocation rule p^M equals the probability that there are at most $k-1$ others with types above t_i. Using $c_\varphi = c_\varphi^{aud}$ on T^k and Lemma 6, this yields

$$\mathbb{E}_{\boldsymbol{t}_{-i}}[p_i^M(t_i, \boldsymbol{t}_{-i})] = \sum_{j=0}^{k-1} \binom{n-1}{j} t_i^{n-1-j}(1-t_i)^j = -\frac{1}{n} c'_\varphi(t_i) - \varphi.$$

Finally, if $t_i \in T^\ell$ we get

$$\mathbb{E}_{\boldsymbol{t}_{-i}}[p_i^M(t_i, \boldsymbol{t}_{-i})] = 0 = -\frac{1}{n} c'_\varphi(t_i) - \varphi.$$

□

Lemma 9. *There is an ex-post allocation rule $p^\ell : \boldsymbol{T} \times [0,1] \to \mathbb{R}^n$ allocating the objects not already allocated under p^M that induces the interim rule $\varphi \mathbf{1}_{T^k \cup T^\ell}$.*

Proof. By Lemma 4, applied with $h(\boldsymbol{t}) = m - \sum_i p_i^M(\boldsymbol{t})$ and $J(\boldsymbol{t}) = \{i \in N : p_i^M(\boldsymbol{t}) = 0\}$, which are symmetric, it is sufficient to show that, for every Borel set $E \subseteq T$,

$$n\varphi \int_E \mathbf{1}_{T^k \cup T^\ell}(t)\,\mathrm{d}t \leqslant \int \min\Big\{ \sum_{i \in I(\boldsymbol{t},E)} \big(1 - p_i^M(\boldsymbol{t})\big), m - \sum_i p_i^M(\boldsymbol{t}) \Big\}\,\mathrm{d}\boldsymbol{t}. \tag{16}$$

Since the right-hand side is increasing in E and the left-hand side only depends on $E \cap [0, x_m]$, we can assume $E \subseteq [0, x_m]$. Let $F := [0, x_m] \backslash E$ and let z denote the Lebesgue measure of F, so that the left-hand side of (16) equals $n\varphi(x_m - z)$.

We claim that, at every type profile $\boldsymbol{t}$,

$$\begin{aligned} &\min\Big\{ \sum_{i \in I(\boldsymbol{t},E)} \big(1 - p_i^M(\boldsymbol{t})\big),\ m - \sum_i p_i^M(\boldsymbol{t}) \Big\} \\ &\quad = m - \sum_i p_i^M(\boldsymbol{t}) - \max\Big\{ \sum_{i \in I(\boldsymbol{t},F)} \big(1 - p_i^M(\boldsymbol{t})\big) - (n-m), 0 \Big\}. \end{aligned}$$

The maximum on the right-hand side is the number of remaining objects in excess of the agents in E who do not obtain an object under p^M. If $\sum_i p_i^M(\boldsymbol{t}) = m$, both sides are zero, since at most $n-m$ agents do not obtain an object. If $\sum_i p_i^M(\boldsymbol{t}) < m$, then fewer than m agents have types in T^m, and all of them obtain an object. Hence, every agent who does not obtain an object has a type in E or in F, and the left-hand side equals

$$\min\Big\{ n - \sum_i p_i^M(\boldsymbol{t}) - \sum_{i \in I(\boldsymbol{t},F)} \big(1 - p_i^M(\boldsymbol{t})\big),\ m - \sum_i p_i^M(\boldsymbol{t}) \Big\}.$$

Since the first argument of the minimum equals the second plus $(n-m) - \sum_{i \in I(\boldsymbol{t},F)}(1 - p_i^M(\boldsymbol{t}))$, this minimum equals the right-hand side. Moreover, since $c_\varphi(0) = \min\{m, k + n\varphi\} = m$ and $c_\varphi(1) = 0$, Lemma 8 implies $\int \sum_i p_i^M(\boldsymbol{t})\,\mathrm{d}\boldsymbol{t} = m - n\varphi x_m$. Therefore, (16) is equivalent to

$$\int \max\Big\{ \sum_{i \in I(\boldsymbol{t},F)} \big(1 - p_i^M(\boldsymbol{t})\big) - (n-m), 0 \Big\}\,\mathrm{d}\boldsymbol{t} \leqslant n\varphi z. \tag{17}$$

We next show that the left-hand side of (17) is largest when $F = [0, z]$. Rearrange F into $[0, z]$ and E into $(z, x_m]$, preserving Lebesgue measure and the order within each set, and leave types above x_m unchanged. Applying this rearrangement to each agent's type yields a type profile that is again uniformly distributed on $\boldsymbol{T}$.

An agent with a type in F moves weakly downwards. After the rearrangement, all agents with types outside F lie above him, and the order among agents with types in F is preserved. Since an agent with a type in $[0, x_m]$ obtains an object under p^M only if his type exceeds x_k and is among the k highest, an agent with a type in F who does not obtain an object still does not obtain one after the rearrangement. Thus, at almost every type profile, the integrand in (17) is at most the integrand for $F = [0, z]$ at the rearranged profile. Since the rearranged profile has the same distribution, the claim follows.

It remains to show (17) for $F = [0, z]$, $z \in [0, x_m]$. We first show that, at every type profile, the integrand in (17) equals the number of agents with types in $[0, z]$ whose types are among the m highest but who do not obtain an object under p^M. If at least m agents have types above z, both numbers are zero. Otherwise, the $n - m$ agents whose types are not among the m highest have types in $[0, z]$, and none of them obtains an object under p^M. Hence, the number of agents with types in $[0, z]$ who do not obtain an object exceeds $n - m$ by exactly the number of such agents whose types are among the m highest.

It follows that the left-hand side of (17) equals the expected number of agents with types in $[0, z]$ whose types are among the m highest, minus the expected number of those who obtain an object under p^M. The first expectation is $m - c^{allo}(z)$, since $c^{allo}(z)$ is the expected number of the m highest types that lie above z. An agent with a type in $[0, z]$ obtains an object under p^M only if his type is among the k highest, and hence among the m highest. Therefore, the second expectation equals the expected number of agents with types in $[0, z]$ who obtain an object under p^M, which is

$$n \int_0^z \mathbb{E}_{\boldsymbol{t}_{-i}}[p_i^M(t, \boldsymbol{t}_{-i})] \, \mathrm{d}t = m - c_\varphi(z) - n\varphi z$$

by Lemma 8 and $c_\varphi(0) = m$. Hence, the left-hand side of (17) equals $n\varphi z - \big(c^{allo}(z) - c_\varphi(z)\big) \leqslant n\varphi z$, since $c_\varphi \leqslant c^{allo}$. □

We let $p = p^\ell + p^M$ and denote by P its interim allocation rule. Note that $P(t) = -\frac{1}{n} c'_\varphi(t)$ for all t. Since c_φ is concave, P is nondecreasing, and since $P(0) = \varphi$, we have $P \geqslant \varphi$; in particular, $\varphi = \inf_t P(t)$. Moreover,

$$n \int_0^1 P(t) \, \mathrm{d}t = c_\varphi(0) - c_\varphi(1) = m.$$

Since the allocation rule assigns at most m objects at every type profile, it follows that it assigns all m objects almost surely.

A.5 Feasibility of the verification rule

Define $A := P - \varphi$.

Lemma 10. *The interim audit rule A is feasible given p.*

Proof. Since $p_i^M(\boldsymbol{t}) \leqslant p_i(\boldsymbol{t}, s)$ for every i, $\boldsymbol{t}$, and s, it suffices to show that A is feasible given p^M. By Lemma 5, it is therefore enough to show that, for every Borel set $E \subseteq T$,

$$n \int_E A(t)\,\mathrm{d}t \leqslant \int \min\{|J^{p^M}(\boldsymbol{t}, E)|, k\}\,\mathrm{d}\boldsymbol{t}. \tag{18}$$

Fix E and let $x \in [x_m, 1]$ be such that $E \cap T^m$ has Lebesgue measure $1-x$. By Lemma 8, $A(t) = \mathbb{E}_{\boldsymbol{t}_{-i}}[p_i^M(t, \boldsymbol{t}_{-i})]$ for $t \in T^\ell \cup T^k$ and $A(t) = \mathbb{E}_{\boldsymbol{t}_{-i}}[p_i^M(t, \boldsymbol{t}_{-i})] - \varphi$ for $t \in T^m$. Hence, by symmetry of p^M, $n\int_E A(t)\,\mathrm{d}t = \int |J^{p^M}(\boldsymbol{t}, E)|\,\mathrm{d}\boldsymbol{t} - n\varphi(1-x)$, and (18) is equivalent to

$$\int \max\left\{|J^{p^M}(\boldsymbol{t}, E)| - k, 0\right\}\mathrm{d}\boldsymbol{t} \leqslant n\varphi(1-x). \tag{19}$$

The left-hand side is the expected number of merit recipients in E exceeding the audit capacity k, while the right-hand side is the reduction in required audits for agents in $E \cap T^m$.

We claim that, at every type profile $\boldsymbol{t}$,

$$\max\left\{|J^{p^M}(\boldsymbol{t}, E)| - k, 0\right\} \leqslant \min\{|I(\boldsymbol{t}, E \cap T^m)|, m\} - \min\{|I(\boldsymbol{t}, E \cap T^m)|, k\}.$$

Since the right-hand side is nonnegative, the claim holds if $|J^{p^M}(\boldsymbol{t}, E)| \leqslant k$. Suppose therefore that $|J^{p^M}(\boldsymbol{t}, E)| > k$. Then no agent with a type in T^k obtains an object under p^M. Thus, all agents in $J^{p^M}(\boldsymbol{t}, E)$ have types in $E \cap T^m$, and since at most m agents obtain an object,

$$k < |J^{p^M}(\boldsymbol{t}, E)| \leqslant \min\{|I(\boldsymbol{t}, E \cap T^m)|, m\}.$$

In particular, $\min\{|I(\boldsymbol{t}, E \cap T^m)|, k\} = k$, so subtracting k from the last inequality proves the claim.

Since $E \cap T^m$ has Lebesgue measure $1-x$, $|I(\boldsymbol{t}, E \cap T^m)|$ has the same distribution as $|I(\boldsymbol{t}, (x, 1])|$. Integrating the claim and using the definitions of c^{allo} and c_φ^{aud}, we obtain

$$\begin{aligned}\int \max\left\{|J^{p^M}(\boldsymbol{t}, E)| - k, 0\right\}\mathrm{d}\boldsymbol{t} &\leqslant \int \Big[\min\{|I(\boldsymbol{t}, (x,1])|, m\} - \min\{|I(\boldsymbol{t}, (x,1])|, k\}\Big]\,\mathrm{d}\boldsymbol{t} \\ &= c^{allo}(x) - c_\varphi^{aud}(x) + n\varphi(1-x) \leqslant n\varphi(1-x),\end{aligned}$$

where the last inequality follows because $x \in [x_m, 1]$ and hence $c^{allo}(x) \leqslant c_\varphi^{aud}(x)$ by Lemma 7. This proves (19). □

A.6 Completing the proof of Theorem 1

We first show that guarantees below $\frac{m-k}{n}$ are never optimal. For $\varphi \leqslant \frac{m-k}{n}$, we have

$$c_\varphi^{aud}(t) - c_\varphi^{ic}(t) \leqslant k + n\varphi - m \leqslant 0,$$

so that c^{ic} is never binding. Since c_φ^{aud} is increasing in φ, c_φ is nondecreasing in φ on $[0, \frac{m-k}{n}]$, and strictly increasing near $t = 0$ because $c_\varphi(0) = k + n\varphi$. Since v is strictly increasing, $\int_0^1 c_\varphi(t)\,\mathrm{d}v(t)$ is strictly increasing in φ on $[0, \frac{m-k}{n}]$.

Now let φ maximize $\int_0^1 c_\varphi(t)\,\mathrm{d}v(t)$ over $[0, \frac{m}{n}]$; $\varphi \geqslant \frac{m-k}{n}$ by the previous paragraph. Let $p = p^M + p^\ell$ be the allocation rule constructed in Section A.4 for this φ. By Lemma 10, there is an audit rule a that is feasible given p with interim rule $A = P - \varphi$. Since $\varphi = \inf_t P(t)$, (p, a) is BIC by Lemma 1. Moreover, $n\int_t^1 P(s)\,\mathrm{d}s = c_\varphi(t)$ for all t, so that, by Lemma 2, the payoff of (p, a) is $\int_0^1 c_\varphi(t)\,\mathrm{d}v(t)$. Any other feasible and BIC mechanism has a guarantee $\tilde{\varphi} \leqslant \frac{m}{n}$, since at most m objects are allocated. By Lemma 2 and the choice of φ, its payoff is at most $\int_0^1 c_{\tilde{\varphi}}(t)\,\mathrm{d}v(t) \leqslant \int_0^1 c_\varphi(t)\,\mathrm{d}v(t)$. Therefore, (p, a) is optimal. By construction, it is a merit-with-guarantee mechanism.

Now let $(\tilde{p}, \tilde{a})$ be an optimal symmetric mechanism with guarantee $\tilde{\varphi}$. By optimality and Lemma 2, $\tilde{\varphi}$ maximizes $\int_0^1 c_\varphi(t)\,\mathrm{d}v(t)$ over $[0, \frac{m}{n}]$, so that $\tilde{\varphi} \geqslant \frac{m-k}{n}$, and (7) is binding everywhere. Let p^M be the merit-based allocation rule corresponding to the cutoffs $x_k(\tilde{\varphi})$ and $x_m(\tilde{\varphi})$ determined by $c_{\tilde{\varphi}}$. On T^m and T^k, binding of the allocation and audit constraints, respectively, implies that the corresponding ex-post feasibility bounds are attained almost surely. Since (4) binds on T^m, we have $\tilde{p}_i = p_i^M$ for $t_i \in T^m$. Since (6) binds on T^k, every report in T^k among the k highest is audited and hence obtains an object. Thus, $p^\ell := \tilde{p} - p^M$ only allocates to agents with $p_i^M = 0$ and reports outside T^m. By Lemma 8, p^ℓ induces the interim rule $\tilde{\varphi}$ on $T^\ell \cup T^k$, and incentive compatibility implies property (iii). Hence, $(\tilde{p}, \tilde{a})$ equals a merit-with-guarantee mechanism almost surely. □

B Additional results

B.1 Proof of Proposition 1

By Lemma 2 and Section A.6, choosing the optimal guarantee amounts to maximizing the principal's payoff per agent,

$$U(\varphi) = \frac{1}{n}\int_0^1 c_\varphi(t)\,\mathrm{d}v(t),$$

over $[0, \frac{m}{n}]$. For each type t, c_φ is concave in φ because it is a minimum of three functions that are each affine in φ. Hence, U is concave in φ.

To compute $U'(\varphi)$, note that a marginal increase in φ changes the binding constraint by $-nt$ on T^ℓ, by $n(1-t)$ on T^k, and leaves it unchanged on T^m. Since v is continuous, the finitely many types at which two constraints are tightest have $\mathrm{d}v$-measure zero. Differentiating under the integral therefore gives

$$U'(\varphi) = -\int_{T^\ell} t \,\mathrm{d}v(t) + \int_{T^k} (1-t)\,\mathrm{d}v(t).$$

By Section A.6, every optimal guarantee is at least $\frac{m-k}{n} > 0$. At $\varphi = \frac{m}{n}$, the incentive constraint binds throughout $(0,1)$, so the left derivative is negative. Hence every optimal guarantee lies in $(0, \frac{m}{n})$.

For any optimal φ, the allocation constraint is strictly the tightest constraint near 1, so $x_m < 1$. Moreover, the incentive constraint lies strictly below the allocation constraint near 0, so $x_m > 0$. If $x_k = 0$, the displayed derivative would be positive; if $x_k = x_m$, it would be negative. Neither is consistent with optimality. Thus all three regions are nondegenerate: $0 < x_k < x_m < 1$. Moreover, since $c^{aud}_\varphi(x) - c^{ic}_\varphi(x)$ is strictly decreasing and equals 0 at $x = x_k$, we get

$$0 < c^{aud}_\varphi(0) - c^{ic}_\varphi(0) = k + n\varphi - m,$$

which implies $\varphi > \frac{m-k}{n}$.

By integration by parts and $v(0) = 0$,

$$\int_0^{x_k} t \,\mathrm{d}v(t) = x_k v(x_k) - \int_0^{x_k} v(t)\,\mathrm{d}t,$$
$$\int_{x_k}^{x_m} (1-t)\,\mathrm{d}v(t) = (1-x_m)v(x_m) - (1-x_k)v(x_k) + \int_{x_k}^{x_m} v(t)\,\mathrm{d}t,$$

so that $U'(\varphi) = 0$ is equivalent to (9). Since U is concave, this condition is also sufficient for optimality. □

B.2 Proof of Proposition 2

Step 1: Convergence of $\frac{1}{n}c_\varphi$ for fixed φ

For any fixed value of φ, we establish the pointwise convergence of each component of c_φ.

For $\frac{1}{n}c^{allo}(t)$: Recall that $-\frac{1}{n}\frac{d}{dt}c^{allo}(t) = \sum_{j=0}^{m-1} \binom{n-1}{j} t^{n-1-j}(1-t)^j$. This sum represents the probability of at most $m-1$ successes in $n-1$ independent Bernoulli trials with success probability $1-t$. For any fixed t, as $n \to \infty$ with $m_n/n \to \alpha$:

- If $t < 1-\alpha$: The expected number of successes $(n-1)(1-t)$ exceeds $m-1$ for n large enough, so by the Strong Law of Large Numbers, this probability converges to 0.

- If $t > 1-\alpha$: The expected number of successes $(n-1)(1-t)$ is less than $m-1$ for n large enough, so this probability converges to 1.

Therefore:

$$-\frac{1}{n}\frac{d}{dt}c^{allo}(t) \to \begin{cases} 0 & \text{if } t < 1-\alpha \\ 1 & \text{if } t > 1-\alpha. \end{cases}$$

Since $\frac{1}{n}c^{allo}(0) = \frac{m_n}{n} \to \alpha$ and the derivative converges, the dominated convergence theorem yields:

$$\frac{1}{n}c^{allo}(t) \to \begin{cases} \alpha & \text{if } t < 1-\alpha \\ \alpha - (t-1+\alpha) & \text{if } t > 1-\alpha. \end{cases}$$

For $\frac{1}{n}c^{aud}_{\varphi}(t)$: By analogous arguments using the fact that $k_n/n \to \beta$:

$$\frac{1}{n}c^{aud}_{\varphi}(t) \to \begin{cases} \beta + \varphi(1-t) & \text{if } t < 1-\beta \\ (1+\varphi)(1-t) & \text{if } t > 1-\beta. \end{cases}$$

For $\frac{1}{n}c^{ic}_{\varphi}(t)$: Since $\frac{1}{n}c^{ic}_{\varphi}(t) = \frac{m_n}{n} - t\varphi$,

$$\frac{1}{n}c^{ic}_{\varphi}(t) \to \alpha - t\varphi$$

Since c^{aud}_{φ} and c^{ic}_{φ} are affine in φ, these limits remain valid if φ is replaced by a sequence converging to φ. If $\varphi \geqslant \alpha - \beta$, then $\alpha \geqslant \alpha - \varphi t$, $\beta + \varphi(1-t) \geqslant \alpha - \varphi t$, and $(1+\varphi)(1-t) \geqslant 1-t$, so that the minimum of the three limits is $\min\{1-t, \alpha - \varphi t\}$.

Step 2: Convergence of the optimal guarantee φ_n

Let φ_n denote the optimal guarantee in a market of size n. We claim that $\varphi_n \to \alpha - \beta$.

First, by Section A.6, we know that $\varphi_n \geqslant \frac{m_n - k_n}{n}$ for all n. Since $\frac{m_n - k_n}{n} \to \alpha - \beta$, we have $\liminf_{n\to\infty} \varphi_n \geqslant \alpha - \beta$. On the other hand, suppose towards contradiction that

$$\limsup_{n\to\infty} \varphi_n > \alpha - \beta.$$

Then there exists a subsequence of (φ_n) that converges to some $\varphi' > \alpha - \beta$. By optimality of φ_n,

$$\frac{1}{n}\int_0^1 c_{\varphi_n}(t)\,\mathrm{d}v(t) \geqslant \frac{1}{n}\int_0^1 c_{\frac{m_n-k_n}{n}}(t)\,\mathrm{d}v(t).$$

By Step 1 and the dominated convergence theorem, along the subsequence the left-hand side converges to $\int_0^1 \min\{1-t, \alpha - \varphi' t\}\,\mathrm{d}v(t)$ and the right-hand side converges to $\int_0^1 \min\{1-t, \alpha - (\alpha-\beta)t\}\,\mathrm{d}v(t)$. For $0 < t < \frac{1-\alpha}{1-\alpha+\beta}$, we have $\alpha - \varphi' t < \alpha - (\alpha-\beta)t \leqslant 1-t$, so the first integrand is strictly smaller than the second for these t and weakly smaller for all t. Since v is strictly increasing, the first limit is strictly smaller than the second, a contradiction, and we conclude that $\varphi_n \to \alpha - \beta$.

Step 3: Convergence of the envelope function $\frac{1}{n}c_{\varphi_n}$

With $\varphi_n \to \alpha - \beta$, the envelope function $\frac{1}{n}c_{\varphi_n}(t) = \frac{1}{n}\min\{c^{allo}(t), c^{aud}_{\varphi_n}(t), c^{ic}_{\varphi_n}(t)\}$ converges as follows.

For $t < \frac{1-\alpha}{1-\alpha+\beta}$, we have

$$\lim \frac{1}{n}c^{allo}(t) > \lim \frac{1}{n}c^{aud}_{\varphi_n}(t) = \lim \frac{1}{n}c^{ic}_{\varphi_n}(t).$$

And for $t > \frac{1-\alpha}{1-\alpha+\beta}$, we have

$$\lim \frac{1}{n}c^{allo}(t) < \lim \frac{1}{n}c^{aud}_{\varphi_n}(t) \leqslant \lim \frac{1}{n}c^{ic}_{\varphi_n}(t).$$

Therefore:

$$\frac{1}{n}c_{\varphi_n}(t) \to \begin{cases} \alpha - (\alpha - \beta)t & \text{if } t < \frac{1-\alpha}{1-\alpha+\beta} \\ \alpha - (t - 1 + \alpha) & \text{if } t > \frac{1-\alpha}{1-\alpha+\beta}. \end{cases}$$

Step 4: Convergence of the allocation rule P_n. Let $t^* = \frac{1-\alpha}{1-\alpha+\beta}$. By the strict comparisons of the limiting constraint functions established in Step 3, for all sufficiently large n, the allocation constraint is not binding at any fixed type $t \in (0, t^*)$, whereas it is binding at any fixed type $t \in (t^*, 1)$.

First, consider a fixed type $t \in (0, t^*)$. By the construction of the merit-with-guarantee mechanism,

$$\varphi_n \leqslant P_n(t) \leqslant \varphi_n + \Pr\big(\mathrm{Bin}(n-1, 1-t) \leqslant k_n - 1\big).$$

Since $t < t^* < 1-\beta$, we have $1-t > \beta$. Hence, by the law of large numbers and $k_n/n \to \beta$, the probability on the right-hand side converges to zero. Together with $\varphi_n \to \alpha - \beta$, this implies

$$P_n(t) \to \alpha - \beta.$$

Next, consider a fixed type $t \in (t^*, 1)$. By the construction of the merit-with-guarantee mechanism,

$$P_n(t) = \Pr\big(\mathrm{Bin}(n-1, 1-t) \leqslant m_n - 1\big).$$

Since $t > t^* > 1 - \alpha$, we have $1 - t < \alpha$. By the law of large numbers and $m_n/n \to \alpha$, this probability converges to one. Thus,

$$P_n(t) \to 1.$$

The conclusions at $t = 0$ and $t = 1$ follow directly from $P_n(0) = \varphi_n$ and $P_n(1) = 1$ for all sufficiently large n.

Finally, since $A_n = P_n - \varphi_n$ and $\varphi_n \to \alpha - \beta$, the claimed convergence of the interim audit rule follows. □

Lemma 11. *For all sufficiently large n, the simple mechanism described after Proposition 2 is Bayesian incentive compatible, and its per-capita payoff converges to the limit of the per-capita payoff of optimal mechanisms.*

Proof. Write m, k, and r for m_n, k_n, and r_n, so that $k \leqslant r \leqslant m$. Selection and audit probabilities depend on a report only through whether it is high. Deviations within either group cannot help, so it suffices to compare a high and a low report. The number H of agents with types above $1 - r/n$ is binomially distributed with mean r; let $\delta := \mathbb{E}[(r - H)^+]$. By symmetry, counting allocations under truthful reporting shows that an agent who reports a high type is selected with probability $\mathbb{E}[\min\{H, r\}]/r = 1 - \delta/r$ and is selected but not audited with probability $\mathbb{E}[(\min\{H, r\} - k)^+]/r$. Some objects may remain unallocated if the number of low reports is smaller than the number of objects remaining after the first round. Hence, the number of objects allocated in the second round is $\min\{n - H, m - \min\{H, r\}\}$. By symmetry, an agent who reports a low type obtains an object with probability $\mathbb{E}[\min\{n - H, m - \min\{H, r\}\}]/(n - r)$, which lies between $\frac{m-r}{n-r}$ and $\frac{m-r+\delta}{n-r}$.

A low type does not gain by reporting a high type, since

$$\frac{\mathbb{E}[(\min\{H, r\} - k)^+]}{r} \leqslant \frac{r - k}{r} \leqslant \frac{m - r}{n - r},$$

where the last inequality follows from $r \leqslant nk/(n - m + k)$. Since $\delta \leqslant \frac{1}{2}\sqrt{\mathrm{Var}(H)} \leqslant \frac{1}{2}\sqrt{r}$ and $r/n \to \frac{\beta}{1-\alpha+\beta}$, the allocation probability of a high report converges to 1 and that of a low report to $\alpha - \beta < 1$. Hence, for all sufficiently large n, a high type does not gain by reporting a low type. Moreover, the cutoff $1 - r/n$ converges to $\frac{1-\alpha}{1-\alpha+\beta}$, so the interim allocation probabilities converge to those in Proposition 2 except at the limiting cutoff, and the claimed payoff convergence follows. □

C Generalizations of Border's theorem

In this appendix, we collect all proofs on characterizations of feasible interim rules. These results are derived from generalizations of the result in Border (1991).

Proof of Lemma 3. This follows immediately from Corollary 4(ii) in Che et al. (2013). □

Lemma 4 extends the characterization of feasible interim rules to settings with type-dependent ex-post constraints. This characterization follows from a recent result in Valenzuela-Stookey (2024).

Proof of Lemma 4. We first prove the result for finite type spaces and then extend the result to continuous type spaces via an approximation argument: Assuming that T is finite, Theorem 2 in Valenzuela-Stookey (2024) establishes the statement in Lemma 4.

The approximation argument used to prove Theorem 5 in Che et al. (2013) can then be adapted to establish the conclusion for our setting with infinitely many types and type-dependent ex-post constraints.

It remains to show the last statement. Suppose $P_1 = \cdots = P_n$ and h and J are symmetric, and define, for $E = E_1 \sqcup \cdots \sqcup E_n$,

$$f(E) := \sum_i \int_{E_i} P_i(t)\,\mathrm{d}t - \int \min\{|J(\boldsymbol{t}) \cap I(\boldsymbol{t}, E)|, h(\boldsymbol{t})\}\,\mathrm{d}\boldsymbol{t}.$$

Then f is symmetric. It is also supermodular: if $X \subseteq Y$ and Z is disjoint from Y, then adding Z increases $|J(\boldsymbol{t}) \cap I(\boldsymbol{t}, \cdot)|$ by the same amount at X as at Y. Since $|J(\boldsymbol{t}) \cap I(\boldsymbol{t}, X)| \leqslant |J(\boldsymbol{t}) \cap I(\boldsymbol{t}, Y)|$ and $\min\{\cdot, h(\boldsymbol{t})\}$ is concave, the integrand $\min\{|J(\boldsymbol{t}) \cap I(\boldsymbol{t}, \cdot)|, h(\boldsymbol{t})\}$ therefore increases by less at Y than at X. Hence, if $f(E) > 0$ for some E, Lemma 12 below yields a Borel set $D \subseteq T$ with $f(D \sqcup \cdots \sqcup D) \geqslant f(E) > 0$. □

We can now use the characterization result on feasible interim allocation rules with type-dependent ex-post constraints to prove the result on feasibility of interim audit rules.

Proof of Lemma 5. An audit rule a is feasible given p if and only if, for all $\boldsymbol{t}$, at most k agents are audited and only agents i with $p_i(\boldsymbol{t}) = 1$ are audited. The claim therefore follows from Lemma 4, applied with $h(\boldsymbol{t}) = k$ and $J(\boldsymbol{t}) = \{i \in N : p_i(\boldsymbol{t}) = 1\}$, which are symmetric because p is symmetric. □

In the proof of Lemma 4, we rely on the following lemma, which uses the same symmetrization principle as the proof of Theorem 7 in Che et al. (2013). To state it, we need to introduce a few pieces of notation.

We say that the disjoint union $D_1 \sqcup \cdots \sqcup D_n$ is *symmetric* if $D_i = D_j$ for all i, j. Let $H := \bigsqcup_{i=1}^n [0,1]$. A function $f : 2^H \to \mathbb{R}$ is *symmetric* if for all $D_1 \sqcup \cdots \sqcup D_n$ and i, j,

$$f(D_1 \sqcup \cdots \sqcup D_i \sqcup \cdots \sqcup D_j \sqcup \cdots \sqcup D_n) = f(D_1 \sqcup \cdots \sqcup D_j \sqcup \cdots \sqcup D_i \sqcup \cdots \sqcup D_n).$$

Lemma 12. *Let $H := \bigsqcup_i [0,1]$ and $\mathcal{B}(H)$ denote the subsets $H_1 \sqcup ... \sqcup H_n$ of H such that H_i is a Borel subset of $[0,1]$ for each i. Let $f : \mathcal{B}(H) \to \mathbb{R}$ be supermodular[15] and symmetric. Then, for every $D_1 \sqcup \cdots \sqcup D_n \in \mathcal{B}(H)$, there is a Borel set $D \subseteq [0,1]$ such that $f(D \sqcup \cdots \sqcup D) \geqslant f(D_1 \sqcup \cdots \sqcup D_n)$.*

Proof. Supermodularity implies $f(X \cup Y) + f(X \cap Y) \geqslant f(X) + f(Y)$ for all $X, Y \in \mathcal{B}(H)$. Let $\mathcal{G}$ denote the family of sets $E_1 \sqcup \cdots \sqcup E_n$ such that each E_i is a union of sets of the

[15] A function $f : \mathcal{B}(H) \to \mathbb{R}$ is supermodular if $X, Y, Z \in \mathcal{B}(H)$, $X \subseteq Y$, and $Z \subseteq H \backslash Y$ imply

$$f(X \cup Z) - f(X) \leqslant f(Y \cup Z) - f(Y).$$

form $\bigcap_{j \in S} D_j$, where S is a nonempty subset of $\{1, \ldots, n\}$. Then $\mathcal{G}$ is finite, contains $D_1 \sqcup \cdots \sqcup D_n$, and is closed under permutations, unions, and intersections. Let M denote the maximum of f on $\mathcal{G}$. If $X, Y \in \mathcal{G}$ are maximizers, then $f(X \cup Y) + f(X \cap Y) \geqslant 2M$, and hence $f(X \cup Y) = M$. Therefore, the union X^* of all maximizers is itself a maximizer. By symmetry of f, every permutation of X^* is also a maximizer and is therefore contained in X^*, so that $X^* = D \sqcup \cdots \sqcup D$ for some Borel set D. Finally, $f(X^*) = M \geqslant f(D_1 \sqcup \cdots \sqcup D_n)$. □

D Revelation Principle

For completeness, we show that it is without loss of generality to restrict attention to Bayesian incentive compatible direct mechanisms. Consider any mechanism and a Bayes–Nash equilibrium of the game it induces, in which every type plays a best response. We define a direct mechanism as follows: given the reports, the principal simulates the original mechanism, assuming that each agent follows the equilibrium strategy of his reported type and that every simulated audit reveals the reported type; the random variable s is used to reproduce any randomization in the mechanism or in the equilibrium strategies. The allocation and audit rules are given by the allocations and audits in the simulation; as in Section 2, an agent whose report is found to be false receives no object. Since the original mechanism allocates at most m objects and audits at most k agents, this indeed defines a direct mechanism. If all agents report truthfully, every audit reveals the reported type, so the direct mechanism reproduces the equilibrium outcome and the principal's expected payoff. Suppose instead that one agent misreports while all others report truthfully. Until this agent is audited, the simulation coincides with the play of the original mechanism when he follows the equilibrium strategy of his reported type. If he is never audited, he therefore obtains an object in the direct mechanism if and only if he obtains one after this deviation in the original mechanism. If he is audited, his lie is detected and he obtains no object, which is the worst outcome for him. Hence a misreport in the direct mechanism is at most as attractive as the corresponding deviation in the original mechanism, which is not profitable in equilibrium, and truthful reporting is optimal.

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